

COMBINATIONS



SYNOPSIS

- A selection that can be formed by taking some or all of a finite set of things (or objects) is called a combination.
- Formation of a combination by taking 'r' elements from a finite set A containing 'n' elements, means picking up r-elements subset of A ($n \geq r$).
- In case of combination, order of the objects is not important.
- The number of combinations of 'n' dissimilar things taken 'r' at a time is equal to the number of 'r' elements subsets of a set containing 'n' elements.
- The number of combinations of 'n' dissimilar things taken 'r' at a time is denoted by

$${}^nC_r \text{ or } C(n, r) \text{ or } C\left(\begin{matrix} n \\ r \end{matrix}\right) \text{ or } \left(\begin{matrix} n \\ r \end{matrix}\right).$$

- $${}^nC_r = \frac{n!}{r!(n-r)!}$$
$$= \frac{{}^np_r}{r!} = \frac{n(n-1)(n-2) \dots (n-r+1)}{1.2.3 \dots r}$$
- $${}^nC_0 = {}^nC_n, {}^nC_1 = {}^nC_{n-1},$$
$${}^nC_2 = {}^nC_{n-2}, \dots, {}^nC_r = {}^nC_{n-r}$$

W.E-1: If there are 12 persons in a party, and if each two of them shake hands with each other, the number of handshakes in the party is

Sol: It is to note that, when two persons shake hands, it is counted as one handshake. The total number of handshakes is same as the number of ways of selecting 2 persons among 12 persons.

$$= {}^{12}C_2 = \frac{12!}{10! \times 2!} = 66$$

- $${}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n = 2^n$$
- $$({}^{2n+1}C_0 + {}^{2n+1}C_1 + \dots + {}^{2n+1}C_n) = 2^{2n}$$
- $$({}^{2n+1}C_{n+1} + {}^{2n+1}C_{n+2} + \dots + {}^{2n+1}C_{2n+1}) = 2^{2n}$$
- $${}^nC_r + {}^nC_{r-1} = {}^{(n+1)}C_r$$

- If ${}^nC_r = {}^nC_s$ then $r = s$ or $r+s = n$.
- $$\frac{{}^nC_r}{{}^{n-1}C_{r-1}} = \frac{n}{r}, \frac{{}^nC_r}{{}^{n-2}C_{r-2}} = \frac{n(n-1)}{r(r-1)}, \text{ etc.}$$
- $$\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r} \quad \Rightarrow \quad \frac{{}^nC_r}{{}^{(n-1)}C_r} = \frac{n}{n-r}$$

W.E-2: The least value of 'n' so that $({}^{n-1}C_3 + {}^{(n-1)}C_4) > {}^nC_3$ is [Eamcet 2011]

Sol. $({}^{n-1}C_3 + {}^{(n-1)}C_4) > {}^nC_3 \Rightarrow {}^nC_4 > {}^nC_3 \Rightarrow n > 4+3$
 $\therefore$ The least value of n is 8.

W.E-3: If

$${}^nC_{r-1} = 330, {}^nC_r = 462, {}^nC_{r+1} = 462 \Rightarrow r =$$

(Eamcet-2013)

Sol.
$$\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{462}{330} \Rightarrow \frac{n-r+1}{r} = \frac{7}{5}$$
$$\Rightarrow 5n - 5r + 5 = 7r \Rightarrow 5n = 12r - 5$$
$$\frac{{}^nC_{r+1}}{{}^nC_r} = \frac{462}{462} \Rightarrow \frac{n-r}{r+1} = 1$$
$$n - r = r + 1 \Rightarrow n = 2r + 1$$
$$5n = 12r - 5 \Rightarrow 5(2r + 1) = 12r - 5$$
$$\Rightarrow 10r + 5 = 12r - 5 \Rightarrow 2r = 10 \Rightarrow r = 5$$

- If nC_r is greatest, then $r = \frac{n}{2}$ when n is even,

$$r = \frac{n-1}{2} \text{ or } \frac{n+1}{2} \text{ when n is odd}$$

➤ **Vandermonde's Theorem :**

$${}^mC_0 \cdot {}^nC_r + {}^mC_1 \cdot {}^nC_{r-1} + {}^mC_2 \cdot {}^nC_{r-2} + \dots$$

$$\dots + {}^mC_{r-1} \cdot {}^nC_1 + {}^mC_r \cdot {}^nC_0 = {}^{m+n}C_r.$$

W.E-4: The sum $\sum_{r=0}^m \binom{10}{i} \binom{20}{m-i}$ is maximum when m is

Sol:
$$\sum_{i=0}^M \binom{10}{i} \binom{20}{m-i} = \sum_{i=0}^m {}^{10}C_i {}^{20}C_{m-i}$$
$${}^{10}C_0 {}^{20}C_m + {}^{10}C_1 {}^{20}C_{m-1} + \dots + {}^{10}C_m {}^{20}C_0 = {}^{30}C_m$$
$${}^{30}C_m \text{ is maximum when } m = \frac{30}{2} = 15$$

- nC_r ($r \neq 0, n$) is divisible by ' n ' only if ' n ' is prime number. As 6C_2 is not divisible by 6 but 7C_4 is divisible by 7.
- The number of combinations of ' n ' things taken ' r ' at a time in which
 - 's' particular things will always occur is ${}^{(n-s)}C_{r-s}$
 - 's' particular things will never occur is ${}^{n-s}C_r$
 - 's' particular things always occur and 'p' particular things never occur is ${}^{(n-p-s)}C_{r-s}$.
- If ' n ' different objects are in a row then the number of ways of selecting ' r ' objects at a time so that no two of these ' r ' objects are consecutive is $(n-r+1)C_r$.
- If ' n ' different objects are on a circle then the number of ways of selecting ' r ' objects at a time so that no two of these ' r ' objects are consecutive is $(n-r+1)C_r - (n-r-1)C_{r-2}$

W.E.5: In how many ways can a cricket eleven be chosen out of a batch of 15 players if
 (a) There is no restriction on the selection
 (b) a particular player is always chosen
 (c) a particular player is never chosen ?

Sol : (i) The total number of ways of selecting 11 players out of 15 is

$$= {}^{15}C_{11} = {}^{15}C_{15-11} = {}^{15}C_4 = \frac{15 \times 14 \times 13 \times 12}{4 \times 3 \times 2 \times 1} = 1365$$

(ii) if a particular player is always chosen. This means that 10 players are selected out of the remaining 14 players. Hence Required number of ways $= {}^{14}C_{10} = {}^{14}C_{14-10} = {}^{14}C_4 = 1001$

(iii) If a particular player is never chosen. This means that 11 players are selected out of the remaining 14 players. Hence Required number of ways $= {}^{14}C_{11} = {}^{14}C_{14-11} = {}^{14}C_3 = 364$

W.E.6: The number of ways in which a team of eleven players can be selected from 22 players always including 2 of them and excluding 4 of them is

Sol : Required number of ways

$${}^{(n-p-s)}C_{r-s} = {}^{22-4-2}C_{11-2} = {}^{16}C_9$$

Groups:

- Number of ways in which $(m+n)$ items can be divided into two unequal groups containing ' m ' and ' n ' items is ${}^{m+n}C_m = \frac{(m+n)!}{m!n!}$
 - Number of ways of distributing $(m+n)$ different things to two persons so that one gets ' m ' things and the other gets ' n ' things is $\frac{(m+n)!}{m!n!} \times 2!$
 - Number of ways of dividing $2n$ different things into two groups, each containing ' n ' things and the order of the groups is not important, is $\frac{(2n)!}{2!(n!)^2}$.
 - Number of ways of dividing $2n$ different things in two groups, each containing ' n ' things and the order of the groups is important, is $\frac{(2n)!}{(n!)^2}$.
 - The number of ways in which ' mn ' different items can be divided equally into ' m ' groups, each containing ' n ' objects and the order of the groups is not important is $\left[\frac{(mn)!}{(n!)^m} \right] \frac{1}{m!}$
 - The number of ways in which ' mn ' different objects can be divided equally into ' m ' groups, each containing ' n ' objects and the order of groups is important, is $\left[\left[\frac{(mn)!}{(n!)^m} \right] \frac{1}{m!} \right] m! = \frac{(mn)!}{(n!)^m}$
 - The number of ways in which $(m+n+p)$ things can be divided into three different groups of m, n and p things respectively is $\frac{(m+n+p)!}{m!n!p!}$
 - The required number of ways of dividing ' $3n$ ' things into three groups of ' n ' each $= \frac{1}{3!} \cdot \frac{(3n)!}{n! \cdot n! \cdot n!}$
- When the order of groups has importance then required number of ways $= (3n)! / (n!)^3$.
- The number of ways in which ' n ' different things can be arranged into ' r ' different groups is ${}^{n+r-1}P_n$ or $n! \cdot {}^{n-1}C_{r-1}$ according to blank groups are admissible or not admissible.

- The number of ways in which 'n' different things can be distributed into 'r' different groups is $r^n - {}^r C_1 (r-1)^n + {}^r C_2 (r-2)^n - \dots + (-1)^{r-1} {}^r C_{r-1}$. Here blank groups are not allowed.
- Distribution of 'n' distinct objects in 'r' different boxes if in any box any number of objects are placed (empty boxes are allowed) Here, each object has 'r' possibilities. Hence, number of ways $= r \times r \times r \times r \dots n \text{ times} = r^n$.

W.E-7: The number of ways can a pack of 52 cards be divided equally among four players in order is

Sol: For the first player we have ${}^{52}C_{13}$ choices, for the second player ${}^{39}C_{13}$ choices, for the third player ${}^{26}C_{13}$ choices and for the last player we have ${}^{13}C_{13}$ choices. Hence, the total number of ways $= {}^{52}C_{13} \times {}^{39}C_{13} \times {}^{26}C_{13} \times {}^{13}C_{13} = \frac{52!}{(13!)^4}$

W.E-8: The number of ways of dividing 20 persons into 10 couples is

Sol: Here the order of the couples is not important. So, required number of ways is $\frac{20!}{(2!)^{10} 10!}$

- If 'n' points are on the circumference of a circle are given, then
 - number of straight lines $= n_{C_2}$.
 - number of triangles $= n_{C_3}$
 - number of quadrilaterals $= n_{C_4}$ so on.
- If a polygon has 'n' sides then the number of diagonals in it is ${}^n C_2 - n$ (or) $\frac{n(n-3)}{2}$.
- In a plane there are 'n' points and no three of which are collinear except 'k' points which lie on a line. Then
 - Number of straight lines that can be formed by joining them $= {}^n C_2 - {}^k C_2 + 1$.
 - Number of triangles that can be formed by joining them $= {}^n C_3 - {}^k C_3$.

- If a set of 'm' parallel lines are intersected by another set of 'n' parallel lines then the number of parallelograms that can be formed $= ({}^m C_2) \cdot ({}^n C_2)$
- Number of rectangles of any size in a square of

$n \times n$ is $\sum_{r=1}^n r^3$ and number of squares of any size is

$$\sum_{r=1}^n r^2.$$

- In a rectangle of $n \times p$ ($n < p$) number of rectangles of any size is $\frac{np}{4} (n+1) (p+1)$ and number of

squares of any size is $\sum_{r=1}^n (n+1-r) (p+1-r)$.

- Number of rectangles on a chess board (including squares) $= 1296$
- Number of squares of all dimensions on a chess board $= 204$
- Number of rectangles on a chess board which are not squares $= 1092$.

W.E-9: Out of thirty points in a plane, eight of them are collinear. The number of straight lines that can be formed by joining these points is (Eamcet - 2014)

Sol. The number of straight lines formed $= {}^{30}C_2 - {}^8C_2 + 1 = 408$

W.E-10: The straight lines l_1, l_2, l_3 are parallel and lie in the same plane. A total number of m points are taken on l_1 , n points on l_2 , k points on l_3 . The maximum number of triangles formed with vertices of these points are

Sol. Total number of points $= m + n + k$
 $\therefore$ Number of selections taking 3 at a time $= {}^{m+n+k}C_3$
 The points on the same line will not give any triangle.
 $\therefore$ Number of triangles $= {}^{m+n+k}C_3 - {}^m C_3 - {}^n C_3 - {}^k C_3$

Problems based on certain theorems on combinations:

- The total number of combinations of $(p_1 + p_2 + \dots + p_k)$ things taken any number at a time when ' p_1 ' things are alike of one kind, p_2 things are alike of second kind, ... p_k things are alike of k^{th} kind, is $(p_1 + 1)(p_2 + 1) \dots (p_k + 1)$.
- The total number of combinations of $(p_1 + p_2 + \dots + p_k)$ things taken one or more at a time when p_1 things are alike of one kind, p_2 things are alike of second kind, ... p_k things are alike of k^{th} kind, is $(p_1 + 1)(p_2 + 1) \dots (p_k + 1) - 1$.
- The total number of combinations of ' n ' different things taken any number at a time is 2^n (or)
- The number of ways of answering **all** of ' n ' questions when each question has an alternative is 2^n .
- The total number of combinations of ' n ' different things taken one or more at a time is $2^n - 1$. (or) The number of ways of answering one or more of ' n ' questions is $2^n - 1$.
- The number of ways of answering one or more of ' n ' questions when each question has an alternative is $3^n - 1$.

W.E-11: *There are 10 lamps in a hall. Each one of them can be switched on independently. The number of ways in which the hall can be illuminated is*

Sol. Required number of ways

$$= {}^{10}C_1 + {}^{10}C_2 + {}^{10}C_3 + \dots + {}^{10}C_{10} = 2^{10} - 1$$

W.E-12: *The number of different sums that can be formed with the coins a rupee, a 50 paise, a 25 paise, a 10 paise, and a 5 paise is*

Sol. One rupee coin can be selected in two ways. That is either selecting or non selecting a coin. Similarly the remaining coins also can be selected in two ways. In this process selecting none of the coins is selected is one way.

The number of different sums is

$$(1+1)(1+1)(1+1)(1+1)(1+1) - 1 = 31$$

W.E-13: *The number of ways in which ' n ' distinct objects can be put into two different boxes is (Each box can hold ' n ' objects)*

Sol: Let the two boxes be B_1 and B_2 . There are two choices for each of the n objects. So, the total number of ways is $2 \times 2 \times \dots \times n \text{ times} = 2^n$

W.E-14: *The number of ways in which ' n ' distinct objects can be put into two different boxes so that no box remains empty, is*

Sol: Each object can be put either in box B_1 (say) or in box B_2 (say). So, there are two choices for each of the ' n ' objects. Therefore the number of choices for ' n ' distinct objects is $2 \times 2 \times \dots$ upto n times $= 2^n$. This includes either the first or the second box being empty. Thus $2^n - 2$ ways.

W.E-15: *The number of ways in which ' n ' distinct objects can be put into two identical boxes so that no box remains empty, is*

Sol: We know that there are $2^n - 2$ ways in which neither box is empty. Since two boxes are identical, required no. of ways $= \frac{1}{2}(2^n - 2)$

➤ **Exponent of 'P' (P is prime) in $n!$ ($n \in \mathbb{N}$)**
(OR) Highest power of P in $n!$ is

$$\left[\frac{n}{P} \right] + \left[\frac{n}{P^2} \right] + \left[\frac{n}{P^3} \right] + \dots$$

Where $[x]$ denotes greatest integer function $\leq x$

W.E-16: *Exponent of 3 in $100!$ is*

Sol: $\left[\frac{100}{3} \right] + \left[\frac{100}{3^2} \right] + \left[\frac{100}{3^3} \right] + \left[\frac{100}{3^4} \right] + \left[\frac{100}{3^5} \right] + \dots$
 $= 33 + 11 + 3 + 1 + 0 + \dots = 48$

➤ If $n! = 2^\alpha 3^\beta 5^\gamma 7^\delta \dots$ where $\alpha > \beta > \gamma \dots$ then $n!$ ends with γ zeros (i.e. that is exponent of 5).

W.E-17:- *No. of zeroes in $100!$ (or) Exponent of 5 in $100!$*

Sol: $\left[\frac{100}{5} \right] + \left[\frac{100}{5^2} \right] + \left[\frac{100}{5^3} \right] + \dots = 20 + 4 + 0 = 24$

➤ Let $N = p_1^{\alpha_1} \cdot p_2^{\alpha_2} \cdot p_3^{\alpha_3} \dots p_k^{\alpha_k}$ where $p_1, p_2, p_3, \dots, p_k$ are different primes and $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_k$ are natural numbers then

➤ The total number of divisors of N
 $= (\alpha_1 + 1)(\alpha_2 + 1) \dots (\alpha_k + 1)$

➤ The total number of proper divisors of N (excluding 1 and N) $= (\alpha_1 + 1)(\alpha_2 + 1) \dots (\alpha_k + 1) - 2$

- The total number of divisors of N excluding either 1 or $N = (\alpha_1 + 1)(\alpha_2 + 1) \dots (\alpha_k + 1) - 1$

W.E-18: The number of non trivial divisors of 2160 is

Sol. $2160 = 2^4 \cdot 3^3 \cdot 5^1$

$$\text{Non-trivial divisors} = (4+1)(3+1)(1+1) - 2 = 38$$

- The sum of all divisors

$$= \left(\frac{p_1^{\alpha_1+1} - 1}{p_1 - 1} \right) \left(\frac{p_2^{\alpha_2+1} - 1}{p_2 - 1} \right) \dots \left(\frac{p_k^{\alpha_k+1} - 1}{p_k - 1} \right)$$

- The number of ways in which N can be resolved as a product of two factors

$$\frac{(\alpha_1+1)(\alpha_2+1) \dots (\alpha_k+1)}{2} \quad (N \text{ is not a perfect square})$$

$$\frac{(\alpha_1+1)(\alpha_2+1) \dots (\alpha_k+1)+1}{2} \quad (N \text{ is a perfect square})$$

W.E-19: The number of even proper divisors of 196 is

Sol. $196 = 2^2 \times 7^2$

$$\therefore \text{No of even proper divisors} = (2)(2+1)-1=5$$

W.E-20: The number of ways in which 900 can be resolved as a product of two factors is

Sol. $900 = 2^2 \times 3^2 \times 5^2$

$$\text{No. of ways} = \frac{(2+1)(2+1)(2+1)+1}{2} = 14$$

- The number of ways in which N can be resolved into two factors which are relatively prime $= 2^{r-1}$ where 'r' is the number of distinct primes.

W.E-21: The number of ways in which 8100 can be split in two factors which are relative prime is

Sol. $8100 = 2^2 \times 3^4 \times 5^2$

$$\text{No. of ways} = 2^{r-1} = 2^{3-1} = 4$$

- If $N = 2^{\alpha_1} \cdot 3^{\alpha_2} \cdot 5^{\alpha_3} \cdot 7^{\alpha_4} \dots$

- The total number of divisors of N are $(\alpha_1 + 1)(\alpha_2 + 1)(\alpha_3 + 1)(\alpha_4 + 1) \dots$
- The total number of odd divisors of N are $(\alpha_2 + 1)(\alpha_3 + 1)(\alpha_4 + 1) \dots$
- The total number of even divisors of N are $\alpha_1(\alpha_2 + 1)(\alpha_3 + 1)(\alpha_4 + 1) \dots$

- d) The sum of all the divisors of N are

$$\left(\frac{2^{\alpha_1+1} - 1}{2 - 1} \right) \left(\frac{3^{\alpha_2+1} - 1}{3 - 1} \right) \left(\frac{5^{\alpha_3+1} - 1}{5 - 1} \right) \dots$$

- e) The sum of all odd divisors of N are

$$\left(\frac{3^{\alpha_2+1} - 1}{3 - 1} \right) \left(\frac{5^{\alpha_3+1} - 1}{5 - 1} \right) \left(\frac{7^{\alpha_4+1} - 1}{7 - 1} \right) \dots$$

- f) The sum of all even divisors of N are (d) - (e).

W.E-22: The sum of odd divisors of 360 is

Sol. $360 = 2^3 \times 3^2 \times 5^1$

$$\text{Sum of odd divisors} = \left(\frac{3^{2+1} - 1}{3 - 1} \right) \left(\frac{5^{1+1} - 1}{5 - 1} \right) = 78$$

- The integral solutions of the equation $\frac{1}{x} + \frac{1}{y} = \frac{1}{n}$;

$$x, y, n \in N \Rightarrow (x-n)(y-n) = n^2$$

$$\Rightarrow x = n + \alpha; y = n + \frac{n^2}{\alpha}$$

where α is a divisor of n^2

The number of positive integral solutions (x, y) is equal to the number of divisors of n^2

W.E-23: The number of integral solutions of

$$x^2 + y^2 = x^2 y^2 \text{ is}$$

Sol. $x = 0, y = 0$ is only one solution

- The number of ways of selecting 3 natural numbers out of

i) $2n$ consecutive natural numbers such that they are in AP $= 2 \binom{n}{2}$

ii) $(2n+1)$ consecutive natural numbers such that they are in A.P $= \binom{n}{2} + (n+1)_{C_2}$

Reason: Selecting any three numbers which are in A.P = Selecting any two numbers whose sum is even

- Number of selections of r things from 'n' things when 'p' particular things are not together in any selection $= {}^n C_r - {}^{n-p} C_{r-p}$

- Number of selections of 'r' consecutive things out of 'n' different things in a row $= n - r + 1$

- Number of selections of 'r' consecutive things out of 'n' different things around a circle

$$= \begin{cases} n, & \text{when } r < n \\ 1, & \text{when } r = n \end{cases}$$

- Number of selections of 'r' things ($r \leq n$) out of 'n' identical things = 1
- Number of selections of none or more things out of 'n' identical things = $n + 1$
- Number of selections of one or more things out of 'n' identical things = n
- Selections when both (identical and distinct) objects are present, the no. of selections taking atleast one out of $P_1 + P_2 + \dots + P_n + S$ objects when P_1 are alike of one kind, P_2 are alike of second kind, so on, P_n are alike of nth kind and S are distinct is
- $$\{(P_1 + 1)(P_2 + 1) \dots (P_n + 1) \cdot 2^S\} - 1$$
- The total number of ways of dividing 'n' identical items among 'r' persons, each one of whom, can receive 0, 1, 2, or more items ($\leq n$) is ${}^{n+r-1}C_{r-1}$ (OR) The total number of ways of dividing 'n' identical objects into 'r' ordered groups, if blank groups are allowed, is ${}^{n+r-1}C_{r-1}$.
- The total number of ways of dividing 'n' identical items among 'r' persons, each one of whom receives at least one item is ${}^{n-1}C_{r-1}$ (OR) The number of ways in which 'n' identical items can be divided into 'r', ordered groups such that blank groups are not allowed, is ${}^{n-1}C_{r-1}$.
- NOTE:** The number of combinations of 'n' things taken 'r' at a time when repetitions are allowed = The number of combinations of $(n + r - 1)$ things taken 'r' at a time when repetitions are not allowed.
- The number of positive integral solutions of the equation $x_1 + x_2 + x_3 + \dots + x_r = n$ is ${}^{n-1}C_{r-1}$.
- The number of non-negative integral solutions of the equation $x_1 + x_2 + x_3 + \dots + x_r = n$ is ${}^{n+r-1}C_{r-1}$.
- Number of non negative integral solutions of $x_1 + x_2 + \dots + x_r \leq n$ is ${}^{(n+r)}C_r$.
- The number of ways in which 'n' identical things can be distributed into 'r' different groups is ${}^{n+r-1}C_{r-1}$ or ${}^{n-1}C_{r-1}$ according as blank groups are allowed or blank groups not allowed.

W.E-24: The number of integers x, y, z, w such that $x + y + z + w = 20$ and $x, y, z, w \geq -1$ is

Sol. Put $x = a - 1, y = b - 1, z = c - 1, w = d - 1$, then $a, b, c, d \geq 0$ and

$$(a - 1) + (b - 1) + (c - 1) + (d - 1) = 20$$

$$\Rightarrow a + b + c + d = 24$$

The number of non negative integral solutions of this equation is ${}^{24+4-1}C_{4-1} = {}^{27}C_3$

$\therefore$ Required no. of solutions = ${}^{27}C_3$

W.E-25: The number of ways in which 12 identical things can go into 5 purses such that no purse being empty is

Sol. $n = 12; r = 5$
No. of positive integral solutions of $x_1 + x_2 + x_3 + x_4 + x_5 = 12$ is ${}^{n-1}C_{r-1} = {}^{11}C_4$

W.E-26: The number of integral solutions of equation $x + y + z + t = 29$, when $x \geq 1, y \geq 2, z \geq 3$ and $t \geq 0$ is

Sol. Coefficient of p^{29} in

$$(p^1 + p^2 + \dots)(p^2 + p^3 + \dots)(p^3 + p^4 + \dots)(p^0 + p^1 + p^2 + \dots)$$

$$x + y + z + t = 29$$

Let $x' = x - 1, y' = y - 2, z' = z - 3, t' = t$
where $x', y', z', t' \geq 0$
 $\Rightarrow x' + y' + z' + t' = 29 - 1 - 2 - 3 = 23$
here $n = 23, r = 4$ ${}^{n+r-1}C_{r-1} = {}^{26}C_3 = 2600$

W.E-27: The number of ways of selecting 10 balls out of an unlimited number of white, red, green and blue balls is

Sol. The number of ways of selecting 10 balls out of an unlimited number of 4 colours balls

$$= \text{Coefficient of } x^{10} \text{ in } (1 + x + x^2 + \dots)^4 =$$

$$\text{Coefficient of } x^{10} \text{ in } (1 - x)^{-4}$$

$$= \text{Coefficient of } x^{10} \text{ in } 1 + {}^4C_1 x + {}^4C_2 x^2 + \dots = {}^{13}C_{10}$$

$$= {}^{13}C_3 = \frac{13 \times 12 \times 11}{6} = 286 \quad (\text{or})$$

Required no. of ways

$$= {}^{n+r-1}C_{r-1} = {}^{13}C_3 = 286 \text{ where } n = 10, r = 4$$

Miscellaneous:

- 'n' st. lines are drawn in the plane such that no two lines are parallel and no three lines are concurrent. Then the no. of parts into which these lines divide the plane is $1 + \frac{n(n+1)}{2}$.
- Number of points of intersections of diagonals of a polygon of 'n' sides which lie completely inside the polygon are nC_4
- There are 'n' straight lines in a plane, no two of which are parallel and no three passes through the same point. Their points of intersection are joined. Then the number of fresh lines thus formed is

$$\frac{1}{8} n(n-1)(n-2)(n-3). \quad (n > 3)$$

Explanation :

For every two lines, there is a point of intersection. $\therefore$ The number of points of intersection is ${}^nC_2 = n(n-1)/2 = m$ (say).

If no three of these m points are collinear, the number of lines formed is mC_2 .

Since every line intersect the other $(n-1)$ lines, every line contains $(n-1)$ points of intersection. Thus there are n sets of $(n-1)$ collinear points. $\therefore n[{}^{(n-1)}C_2]$ lines coincide with the given n lines.

The number of fresh lines formed $= {}^mC_2 - n[{}^{(n-1)}C_2]$

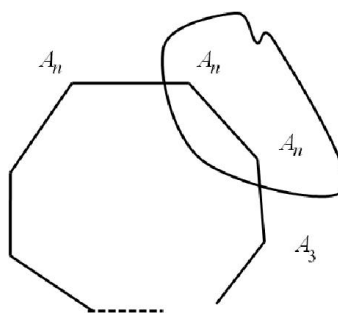
$$\begin{aligned} &= \frac{m(m-1)}{2} - n \frac{(n-1)(n-2)}{2} \\ &= \frac{1}{2} \frac{n(n-1)}{2} \left[\frac{n(n-1)}{2} - 1 \right] - \frac{n(n-1)(n-2)}{2} \\ &= \frac{n(n-1)}{2} \left[\frac{n^2 - n - 2}{4} - n + 2 \right] \\ &= \frac{n(n-1)}{8} [n^2 - n - 2 - 4n + 8] \\ &= \frac{n(n-1)(n^2 - 5n + 6)}{8} \end{aligned}$$

$$= \frac{n(n-1)(n-2)(n-3)}{8}$$

- The number of triangles whose angular points are at the angular points of a given polygon of 'n' sides, but none of whose sides are the sides of the polygon is $\frac{1}{6} n(n-4)(n-5)$.

Explanation :

Let polygon be $A_1 A_2 \dots A_n$.



Number of triangles (i.e., selection of three vertices from n vertices) $= {}^nC_3 = \frac{1}{6} n(n-1)(n-2)$

The number of triangles having only one side common with the polygon $= n(n-4)$

For illustration, consider side $A_1 A_2$, as two points A_3 and A_n are adjacent, 3rd point should be chosen from remaining $(n-4)$ points which can be done in $(n-4)$ ways. Next the number of triangles with two sides common with the polygon. = number of ways of choosing three consecutive vertices

$$(i.e., A_1 A_2 A_3, A_2 A_3 A_4, A_3 A_4 A_5, \dots, A_n A_1 A_2) = n$$

Hence, the required number of triangles none of whose sides are the sides of the polygon

$$\begin{aligned} &= {}^nC_3 - n(n-4) - n \\ &= \frac{1}{6} [n(n-1)(n-2) - 6n(n-4) - 6n] \\ &= \frac{1}{6} n(n-4)(n-5) \end{aligned}$$

- There are ' n ' pairs of shoes in a closet. If $r (< n)$ are selected at random then the number of ways that among the selected shoes
- there is no pair is ${}^nC_r \cdot 2^r$
 - there is atleast one pair is ${}^{2n}C_r - {}^nC_r \cdot 2^r$
 - There is exactly one pair is ${}^nC_1 \cdot {}^{n-1}C_{r-2} \cdot 2^{r-2}$
 - there are exactly 2 pairs is ${}^nC_2 \cdot {}^{n-2}C_{r-4} \cdot 2^{r-4}$

Explanation :

- Treat ' n ' pairs of shoes as ' n ' different units. Select ' r ' out of them in nC_r ways and select one shoe either right or left from each unit in 2C_1 ways. Then the number of ways in which there is no pair from the selected units $= {}^nC_r \times {}^2C_1 \times {}^2C_1 \times \dots \times {}^2C_1$ times ${}^nC_r \times 2^r$
- Number of ways in which there is atleast one pair = Total no. of selections - no. of ways in which there is no pair $= {}^{2n}C_r - {}^nC_r \cdot 2^r$
- Select one pair in nC_1 ways. From the remaining $(n-1)$ units select $(r-2)$ units in ${}^{n-1}C_{r-2}$ ways and select only one shoe (either right or left) in ${}^2C_1 \cdot {}^2C_1 \times \dots \times {}^2C_1$ times. Therefore the number of ways in which there is exactly one pair $= {}^nC_1 \cdot {}^{n-1}C_{r-2} \cdot 2^{r-2}$
- Select two pairs in nC_2 ways. From the remaining $(n-2)$ units, select $(r-4)$ units in ${}^{n-2}C_{r-4}$ and select one shoe either right or left in ${}^2C_1 \times {}^2C_1 \times \dots \times {}^2C_1$ times, so that there is no pair from the remaining units. Hence, the number of ways in which there are exactly two pairs $= {}^nC_2 \cdot {}^{n-2}C_{r-4} \cdot 2^{r-4}$



C.U.Q

1. If $r > 1$ then $\frac{{}^nP_r}{{}^nC_r}$ is
- is an integer
 - may be fraction
 - is an odd number
 - an even number

- The value of $1 \times 3 \times 5 \dots (2n-1) 2^n =$
 1) $\frac{(2n)!}{2^n}$ 2) $\frac{(2n)!}{n!}$ 3) $\frac{n!}{(2n)!}$ 4) $2n$
- If ${}^nC_r + {}^nC_{r+1} = {}^{(n+1)}C_x$ then $x =$
 1) r 2) $r-1$ 3) n 4) $r+1$
- The least value of n so that ${}^nC_6 + {}^nC_7 > {}^{n+1}C_6$ is
 1) 13 2) 12 3) 11 4) 10
- No. of ways of selecting 2 girls and 3 boys from 3 girls and 5 boys is
 1) 20 2) 24 3) 30 4) 48
- A father with 6 children takes 3 at a time to a park without taking the same children. How often father goes to the park?
 1) 14 2) 16 3) 18 4) 20
- From 15 players the number of ways of selecting 6 so as to exclude a particular player is
 1) ${}^{14}C_5$ 2) ${}^{15}C_6$ 3) ${}^{15}C_5$ 4) ${}^{14}C_6$
- The number of three digit numbers of the form xyz where $x > y > z$ is
 1) 120 2) 720 3) 600 4) 100
- A set contains $(2n+1)$ elements. If the number of subsets of this set which contain atmost ' n ' elements is 4096, then the value of n is
 1) 6 2) 15 3) 21 4) 20
- In a library there are $(2n+1)$ books. If a student selects atleast $(n+1)$ books in 256 ways then the number of books in the library is
 1) 7 2) 8 3) 9 4) 6
- A set contains $(2n+1)$ elements. The number of subsets of this set which contains more than ' n ' elements is
 1) 2^{n-1} 2) 2^n 3) 2^{n+1} 4) 2^{2n}
- The number of subsets of the set $A = \{1, 2, 3, \dots, 9\}$ containing at least one odd number is
 1) 324 2) 396 3) 496 4) 512
- The number of ways in which 52 cards can be divided into 4 sets of 13 each is
 1) $\frac{52!}{(13!)^4}$ 2) $\frac{52!}{4!(13!)^4}$ 3) $\frac{52!}{4^{13}}$ 4) $\frac{52!}{13!4^{13}}$
- No. of ways of distributing $(p+q+r)$ different things to 3 persons so that one person gets p things, 2nd person q things 3rd person r things is
 1) $\frac{(p+q+r)!}{p!.q!.r!} \times 3!$ 2) $\frac{(p+q+r)!}{p!.q!.r!}$
 3) $\frac{(p+q+r)!}{3!.p!.q!.r!}$ 4) $p!.q!.r!$

15. The number of quadratic expressions with the coefficients drawn from the set {1, 2, 3, 4} is
1) 27 2) 36 3) 64 4) 48
16. The number of ways in which a man can invite one or more of his 8 friends to dinner is
1) 28 2) 128 3) 240 4) 255
17. The no. of 5 digit numbers that can be made using the digits 1 and 2 and in which atleast one digit is different is
1) 31 2) 32 3) 30 4) 29
18. No. of ways of selecting 3 consecutive objects out of 15 objects (distinct) placed around a circle is
1) ${}^{15}C_r$ 2) ${}^{14}C_3$ 3) 13 4) 15

C.U.Q - KEY

- 1) 4 2) 2 3) 4 4) 1 5) 3 6) 4
7) 4 8) 3 9) 1 10) 3 11) 4 12) 3
13) 2 14) 1 15) 3 16) 4 17) 3 18) 4

C.U.Q - HINTS

1. $\frac{{}^n P_r}{{}^n C_r} = r! = \text{even}$
2. $1.3.5 \dots (2n-1) 2^n$
 $= \frac{1.2.3.4.5 \dots (2n-1) 2n.2^n}{2.4.6. \dots 2n}$
3. ${}^n C_r + {}^n C_{r+1} = {}^{(n+1)} C_{r+1} \Rightarrow x = r + 1$
4. ${}^{n+1} C_7 > {}^{n+1} C_6 \Rightarrow n+1 > 7+6 \Rightarrow n > 12$
 $\Rightarrow n = 13, 14, \dots \therefore n = 13$
5. ${}^3 C_2 \times {}^5 C_3 = 30$
6. No. of times he will go to garden
= No. ways of selecting 3 children from 6
 $= {}^6 C_3 = \frac{6 \times 5 \times 4}{1 \times 2 \times 3} = 20$
7. ${}^{14} C_6 = {}^{14} C_8$
8. Any selection of three digits from the ten digits 0, 1, 2, 3, ..., 9 gives one number.
It is of ${}^{10} C_3$ ways
9. ${}^{(2n+1)} C_0 + {}^{(2n+1)} C_1 + \dots + {}^{(2n+1)} C_n = 4096$
 $\Rightarrow 2^{2n} = 4096 = 2^{12} \Rightarrow n = 6$
10. ${}^{2n+1} C_{n+1} + {}^{2n+1} C_{n+2} + \dots + {}^{2n+1} C_{2n+1} = 256$
Verify

11. ${}^{(2n+1)} C_{n+1} + {}^{(2n+1)} C_{n+2} + \dots + {}^{(2n+1)} C_{2n+1}$
 $= \frac{2^{2n+1}}{2} = 2^{2n}$
12. The total number of subsets = 2^9
The number of subsets without odd numbers = 2^4
 $\therefore$ Required number of subsets
 $= 2^9 - 2^4 = 512 - 16 = 496$
13. $\frac{52!}{(13!)^4 \cdot 4!}$
14. From synopsis $\frac{(p+q+r)!}{p! \cdot q! \cdot r!} \times 3!$
15. $4^3 = 64$
16. ${}^8 C_1 + {}^8 C_2 + \dots + {}^8 C_8 = 2^8 - 1 = 255$
17. $2^5 - 2 = 30$
18. No. of ways of selecting r ($r < n$) consecutive objects from n distinct objects around the circle is 'n'



LEVEL-I (C.W)

PROBLEMS ON ${}^n C_r$ MODEL

1. If ${}^{15} C_{3r} = {}^{15} C_{r+3}$ then $r =$
1) $\frac{3}{2}$ 2) 3 3) 4 4) 5
2. If ${}^{2n} C_3 : {}^n C_2 = 44 : 3$ then $n =$
1) 6 2) 7 3) 8 4) 9
3. If ${}^{n+2} C_8 : {}^{n-2} P_4 = 57 : 16$ then $n =$
1) 17 2) 18 3) 19 4) 20
4. ${}^{14} C_4 + \sum_{j=1}^4 (18-j) C_3 =$
1) ${}^{14} C_5$ 2) ${}^{18} C_5$ 3) ${}^{18} C_4$ 4) ${}^{19} C_4$
5. If ${}^n P_r = 30240$ and ${}^n C_r = 252$ then the ordered pair $(n, r) =$ [EAMCET-2008]
1) (12, 6) 2) (10, 5) 3) (9, 4) 4) (16, 7)
6. If $C(2n, 3) : C(n, 2) = 12 : 1$, then $n =$
1) 4 2) 5 3) 6 4) 8

SELECTION OF DISSIMILAR THINGS

7. A group contains 6 men and 3 women. A committee is to be formed with 5 people containing 3 men and 2 women. The number of different committees that can be formed is
1) 9C_5 2) ${}^6C_3 \times {}^3C_2$ 3) 6C_3 4) 3C_2
8. A team of 11 players has to be chosen from the groups consisting of 6 and 8 players respectively. The number of ways of selecting them so that each selection contains atleast 4 players from the first group is
1) 120 2) 280 3) 344 4) 244
9. A student is to answer 10 out of 13 questions in an examination such that he must choose at least 4 from the first five questions. The number of choices available to him is
1) 196 2) 280 3) 346 4) 140
10. Ten students are participating in a race. The number of ways in which the first three places can be taken as
1) 3 2) ${}^{10}C_3$ 3) ${}^{10}P_3$ 4) ${}^{10}P_4$
11. In a chess tournament, where the participants were to play one game with another. Two chess players fell ill, having played 3 games each. If the total number of games played is 84, the number of participants at the beginning was
1) 15 2) 16 3) 20 4) 21
12. From a company of 20 soldiers any 5 are placed on guard, each batch to watch 5 hours. For what length of time in hours can different batches be selected is
1) ${}^{29}C_5$ 2) ${}^{20}P_5$ 3) ${}^{20}C_5 \times 5$ 4) ${}^{20}P_5 \times 5$
13. 10 persons are seated at round table. The number of ways of selecting 3 persons out of them if no two persons are adjacent to each other is
1) 50 2) 62 3) 56 4) 57

PROBLEMS ON INCLUDING & EXCLUDING

14. The number of ways in which a team of 11 players can be selected from 22 players including 2 of them and excluding 4 of them is
1) ${}^{16}C_6$ 2) ${}^{16}C_7$ 3) ${}^{16}C_8$ 4) ${}^{20}C_7$
15. The number of ways that a volley ball 6 can be selected out of 10 players so that 2 particular players are included is
1) 72 2) 70 3) 68 4) 66

16. In a shelf there are 10 English and 8 Telugu books. The number of ways in which 6 books can be chosen if a particular English book is excluded and a particular Telugu book is excluded is
1) ${}^9C_3 \cdot {}^7C_3$ 2) ${}^{16}C_6$ 3) ${}^9C_3 \cdot {}^8C_3$ 4) ${}^{18}C_8$
17. The number of permutations of n things taken r at a time if 3 particular things always occur is
1) $\frac{(n-3)!}{(n-r)!} r(r-1)(r-2)$ 2) $\frac{(n-3)!}{(r-3)!}$
3) $\frac{(n-3)!}{(n-r)!} \times 3$ 4) $\frac{(n-3)!}{(r-2)!}$
18. The number of all three element subsets of the set $\{a_1, a_2, a_3, \dots, a_n\}$ which contain a_3 is
1) nC_3 2) ${}^{n-1}C_3$ 3) ${}^{n-1}C_2$ 4) nC_2

GEOMETRICAL APPLICATIONS

19. There are 10 straight lines in a plane no two of which are parallel and no three are concurrent. The points of intersection are joined, then the no. of fresh lines formed are
1) 630 2) 615 3) 730 4) 600
20. If a line segment be cut at 'n' points, then the number of line segments formed is
1) $n(n+3)$ 2) $\frac{n(n-3)}{2}$ 3) $\frac{(n+2)(n+1)}{2}$ 4) n
21. The sides AB, BC, CA of a triangle ABC have 3, 4 and 5 interior points respectively on them. The number of triangles that can be formed using these points as vertices is
1) 205 2) 220 3) 225 4) 230
22. There are 'p' points in space of which 'q' points are coplanar. Then the number of planes formed is
1) ${}^pC_3 - {}^qC_3$ 2) ${}^pC_3 - {}^qC_3 + 1$
3) ${}^pC_2 - {}^qC_2$ 4) ${}^pC_3 - {}^qC_2$
23. In a plane there are 37 straight lines of which 13 pass through the point A and 11 pass through the point B. Besides, no three lines pass through one point, no line passes through both point A and B, and no two are parallel. Then the number of points of intersection of the lines is
1) 535 2) 601 3) 728 4) 600

24. A regular polygon of n sides has 170 diagonals. Then $n =$ [EAM-2012]
 1) 12 2) 17 3) 20 4) 25
25. If m parallel lines in a plane are intersected by n parallel lines then number of parallelograms formed is
 1) $\frac{m!n!}{(2!)^2}$ 2) $\frac{m!n!}{(m-2)!(n-2)!}$
 3) $\frac{m!n!}{(2!)^2(m-2)!(n-2)!}$ 4) $\frac{(m+n)!}{(m+n-2)!2!}$
26. The number of rectangles which are not squares in a chess board is
 1) ${}^8C_2 \times {}^8C_2 - 8^2$ 2) $(8_{C_2} \times 8_{C_2}) - \sum 8^2$
 3) ${}^9C_2 \times {}^9C_2 - 8^2$ 4) $(9_{C_2} \times 9_{C_2}) - \sum 8^2$
27. The number of ways of selecting two squares (1×1) in a chess board such that they have a side in common is
 1) 224 2) 112 3) 56 4) 68

TOTAL NUMBER OF COMBINATIONS

28. A basket contains 4 Oranges, 5 Apples, 6 Mangoes. The number of ways a person make selection of fruits from the basket is
 1) 209 2) 210 3) 211 4) 212
29. No. of ways of selecting none or more of 10 identical things is
 1) 1 2) 10 3) 11 4) 9
30. The no of ways of selecting atleast one letter from the letters of the word "PROPORTION" is.
 1) 288 2) 286 3) 289 4) 287
31. The number of different products that can be formed with 8 prime numbers is
 1) 247 2) 252 3) 5 4) 248
32. At an election, a voter may vote for any number of candidates, not greater than the number to be elected. There are 10 candidates and 4 are to be elected. If a voter votes for atleast one candidate, then the number of ways in which he can vote, is
 1) 385 2) 1110 3) 5040 4) 6210

DISTRIBUTION OF DISSIMILAR THINGS INTO GROUPS

33. No. of ways of dividing 80 cards into 5 equal groups of 16 each is
 1) $\frac{80!}{(16!)^5}$ 2) $\frac{80!}{(5!)^{16}}$ 3) $\frac{80!}{(5!)^5}$ 4) $\frac{80!}{(16!)^5} \cdot 5!$
34. The number of ways a pack of 52 cards can be divided among four players in 4 sets, three of them having 17 cards each and the fourth one just 1 card is
 1) $\frac{52!}{(17!)^3} \times 4$ 2) $\frac{52!}{3 \cdot (17!)^3}$
 3) $\frac{52!}{3!(17!)^3}$ 4) $\frac{52!}{3^3(17!)^3}$
35. The number of ways can 5 things be divided between A and B so that each receive at least one thing is
 1) 30 2) 60 3) 20 4) 80

DISTRIBUTION OF SIMILAR THINGS INTO GROUPS

36. In how many ways can 3 sovereigns be given away when there are 4 applicants and any applicant may have either 0, 1, 2 or 3 sovereigns?
 1) 15 2) 20 3) 24 4) 48
37. The number of ways of distributing 15 things to 4 persons each receiving atleast two is
 1) 120 2) 60 3) 28 4) 108
38. The number of non-negative integral solutions of $x_1 + x_2 + x_3 + x_4 \leq n$ (where n is a non negative integer) is
 1) ${}^{(n+3)}C_3$ 2) ${}^{(n+5)}C_5$ 3) ${}^{(n+4)}C_4$ 4) ${}^{(n+6)}C_6$

NUMBER OF DIVISORS

39. The number of odd divisors of 128 is
 1) 8 2) 7 3) 0 4) 1
40. The number of odd proper positive divisors of $3^a 6^b 21^c$ is $(a, b, c \in \mathbb{N})$
 1) $(a+1)(b+1)(c+1) - 2$
 2) $(a+b+c+1)(c+1) - 1$
 3) $(a+1)(b+1)(c+1) - 1$
 4) $(a+1)(b+1)(c+1)$

41. The number of ways of writing 98 as the product of two positive integers is

1) 1 2) 2 3) 3 4) 4

APPLICATIONS OF ONTO FUNCTION

42. 5 balls of different colours are to be kept in 3 boxes of different sizes. Each box can hold all five balls. Number of ways in which the balls can be kept in the boxes so that no box remain empty is

1) 60 2) 90 3) 150 4) 200

LEVEL-I (C.W.) - KEY

1) 2	2) 1	3) 3	4) 3	5) 2	6) 2	7) 2
8) 3	9) 1	10) 3	11) 1	12) 3	13) 1	14) 2
15) 2	16) 2	17) 1	18) 3	19) 1	20) 3	21) 1
22) 2	23) 1	24) 3	25) 3	26) 4	27) 2	28) 1
29) 3	30) 4	31) 1	32) 1	33) 4	34) 1	35) 1
36) 2	37) 1	38) 3	39) 4	40) 2	41) 3	42) 3

LEVEL-I (C.W.) - HINTS

- ${}^{15}C_{3r} = {}^{15}C_{r+3} \Rightarrow 3r + (r+3) = 15 \Rightarrow r = 3$
- Use $\frac{2n(2n-1)(2n-2)}{6} \times \frac{2}{n(n-1)} = \frac{44}{3}$
- Use ${}^nC_r = \frac{n!}{r!(n-r)!}$
- ${}^nC_r \times \sum_{i=1}^m (m+n-i)C_{r-1} = {}^{(m+n)}C_r = {}^{18}C_4$
- $r! = \frac{{}^nP_r}{{}^nC_r} = \frac{30240}{252} = 120 = 5!$
 ${}^nP_5 = 30240 = {}^{10}P_5 \therefore (n, r) = (10, 5)$
- $\frac{{}^{2n}C_3}{{}^nC_2} = \frac{12}{1} \left(\text{use } {}^nC_r = \frac{n!}{r!(n-r)!} \right) \Rightarrow n = 5$
- ${}^6C_3 \times {}^3C_2$
- | | |
|------|------|
| A(6) | B(8) |
| 4 | 7 |
| 5 | 6 |
| 6 | 5 |

$${}^6C_4 \times {}^8C_7 + {}^6C_5 \times {}^8C_6 + {}^6C_6 \times {}^8C_5$$

$$= 120 + 168 + 56 = 344$$

- | | |
|---|---|
| A | B |
| 5 | 8 |
| 5 | 5 |
| 4 | 6 |

$${}^5C_5 \cdot {}^8C_5 + {}^5C_4 \cdot {}^8C_6 = 196$$
- $10_{C_3} \times 3!$
- ${}^{(n-2)}C_2 + 6 = 84$
- Number of batches = ${}^{20}C_5$
each batch $\rightarrow$ 5 hours ; Total ${}^{20}C_5 \times 5$
- Use $\frac{n(n-4)(n-5)}{6}$
- ${}^{(22-2-4)}C_{11-2} = {}^{16}C_9 = {}^{16}C_7$
- ${}^8C_4 = 70$
- ${}^{16}C_6$
- Selection of (r-3) things from (n-3) and arrange r things
- Since each set must contain a_3 , other two elements taken from remaining (n-1) can be selected in ${}^{n-1}C_2$ ways
- $\frac{n(n-1)(n-2)(n-3)}{8}$ put $n = 10$
- Total no. of points = 2 ends + n points = n+2 points
- ${}^{12}C_3 - ({}^3C_3 + {}^4C_3 + {}^5C_3)$
- Total - Coplanar + 1
- ${}^{37}C_2 - {}^{13}C_2 - {}^{11}C_2 + 2 = 535$
- $\frac{n(n-3)}{2} = 170$
- ${}^mC_2 \cdot {}^nC_2$
- Synopsis : $({}^9C_2 \times {}^9C_2) - \sum (8)^2$
- $\frac{1}{2} (4 \times 2 + 24 \times 3 + 36 \times 4) = 112$
- $(4+1)(5+1)(6+1) - 1$
- No. of ways of selecting none or more of n identical things is n+1
- PROTIN

PRO

O

No. of ways = $3.3.4.2.2.2.1 = 287$

$((p+1)(q+1) \dots - 1)$
- ${}^8C_2 + {}^8C_3 + {}^8C_4 + {}^8C_5 + {}^8C_6 + {}^8C_7 + {}^8C_8$

32. The desired number is

$$\binom{10}{1} + \binom{10}{2} + \binom{10}{3} + \binom{10}{4}$$

$$= 10 + 45 + 120 + 210 = 385$$
33. $\frac{80!}{(16!)^5 \cdot 5!}$
34. $\frac{52!}{(17!)^3} \times 4$ (Between person we don't write 3! in Dr)
35. ${}^5C_1 + {}^5C_2 + {}^5C_3 + {}^5C_4$
36. $x + y + z + w = 3$
 No. of non-negative integral solutions ${}^{3+4-1}C_{4-1} = 20$
37. $x_1 + x_2 + x_3 + x_4 = 15 - 8 = 7$
 $\therefore$ No. of non-negative integral solutions
 ${}^{7+4-1}C_{4-1} = 120$
38. From synopsis ${}^{n+r}C_r = {}^{n+4}C_4$
39. $128 = 2^7$
40. $3^a 6^b 21^c = 2^b 3^{a+b+c} 7^c$
 $\Rightarrow$ Number of odd proper divisors
 $= (a+b+c+1)(c+1) - 1$, (excluding 1)
41. $98 = 2^1 7^2 \Rightarrow \frac{(1+1)(2+1)}{2} = 3$
42. Boxes
- | | A | B | C |
|---|---|---|---|
| 1 | 1 | 1 | 3 |
| 1 | 3 | 1 | 1 |
| 3 | 1 | 1 | 1 |
| 1 | 2 | 2 | 2 |
| 2 | 1 | 2 | 2 |
| 2 | 2 | 1 | 1 |
- $3 \times {}^5C_3 \times {}^2C_1 \times {}^1C_1 + 3 \times {}^5C_2 \times {}^3C_2 \times {}^1C_1 = 150$ (or)
 $n^m - {}^n C_1 (n-1)^m + {}^n C_2 (n-2)^m - \dots$
 where $n=3, m=5$



LEVEL-I (H.W)

PROBLEMS ON ${}^n C_r$ MODEL

- If ${}^n C_3 = {}^n C_9$ then ${}^n C_2 =$
 1) 66 2) 132 3) 72 4) 98
- If ${}^n C_3 : {}^{2n-1} C_2 = 8 : 15$ then 'n' is
 1) 5 2) 6 3) 8 4) 7
- If ${}^n C_{r-1} = 36, {}^n C_r = 84, {}^n C_{r+1} = 126$ then (n, r) =
 1) (9, 6) 2) (9, 5) 3) (9, 3) 4) (9, 2)
- The least value of the natural number 'n' satisfying $c(n, 5) + c(n, 6) > c(n+1, 5)$
 1) 10 2) 12 3) 13 4) 11

5. ${}^{22}C_5 + \sum_{i=1}^4 ({}^{26-i}C_4) =$
 1) ${}^{27}C_5$ 2) ${}^{27}C_4$ 3) ${}^{26}C_4$ 4) ${}^{26}C_5$
6. If n and r are integers such that $1 < r < n$ then $n \cdot C(n-1, r-1) =$
 1) $C(n, r)$ 2) $n \cdot C(n, r)$
 3) $r C(n, r)$ 4) $(n-1) \cdot C(n, r)$

SELECTION OF DISSIMILAR THINGS

- A committee of 5 is to be formed from 6 boys and 5 girls. The number of ways that the committee can be formed so that the committee contains atleast one boy and one girl having majority of boys is
 1) 240 2) 245 3) 250 4) 275
- The number of different ways in which a committee of 4 members formed out of 6 Asians, 3 Europeans and 4 Americans if the committee is to have atleast one from each of the 3 regional groups is
 1) 320 2) 340 3) 360 4) 380
- 153 games were played at a chess tournament with each contestant playing once against each of the others. The number of participants is
 1) 16 2) 17 3) 18 4) 19
- There were two women participating in a chess tournament. Every participant played two games with the other participants. the number of games that the men played between themselves proved to exceed by 66 the number of games that the men played with the women. The number of participants is
 1) 6 2) 11 3) 13 4) 10
- In how many ways can 12 boys be seated on two benches x, y; 6 on each bench if two of them A, B are to sit on bench x and C, D on the bench y
 1) ${}^8C_4 \cdot 6! \cdot 6!$ 2) ${}^8C_6 \cdot 6! \cdot 6!$ 3) $6! \cdot 6!$ 4) ${}^8C_4 \cdot 6!$
- If 'n' different things are arranged around a circle then the number of ways of selecting 3 objects when no two selected objects are together

- $\frac{(n-4)(n-5)}{3!}$
- $\frac{n(n-4)(n-5)}{3!}$
- $\frac{n(n-5)}{6}$
- $\frac{n(n-4)}{3}$

PROBLEMS ON INCLUDING & EXCLUDING

13. Out of 7 men and 4 women a committee of 5 is to be formed. The number of ways in which this can be done so as to include atleast 2 women is
1) 210 2) 301 3) 294 4) 84
14. A box contains 2 white, 3 black and 4 red balls. (Balls are of different sizes). In how many ways can 3 balls be drawn from the box if atleast one black ball is to be included in the draw is
1) 84 2) 64 3) 60 4) 120
15. The number of ways that a volleyball team of 6 players can be selected out of 10 players so that 2 particular players are excluded is
1) 56 2) 55 3) 27 4) 28
16. There are 10 balls of different colours. In how many ways is it possible to select 7 of them so as to exclude the white and the black ball ?
1) 8 2) 7 3) 16 4) 20
17. If 20% of three element subsets (i.e. subsets containing exactly three elements) of the set $A = \{a_1, a_2, \dots, a_n\}$ contain a_1 , then value of 'n' is
1) 15 2) 16 3) 17 4) 18

GEOMETRICAL APPLICATIONS

18. There are 15 lines terminating at a point. The number of angles formed is
1) 105 2) 120 3) 125 4) 120
19. Given five line segments of length 2, 3, 4, 5, 6 units. Then the number of triangles that can be formed by joining these lines is
1) ${}^5C_3 - 3$ 2) ${}^5C_3 - 1$ 3) 5C_3 4) ${}^5C_3 - 2$
20. The number of triangles whose vertices are at the vertices of an octagon but none of whose sides happen to come from the sides of the octagon is
1. 24 2. 52 3. 48 4. 16
21. If a polygon of n sides has 275 diagonals, then n = [EAM-2007]
1) 25 2) 35 3) 20 4) 15
22. A parallelogram is cut by two sets of m lines parallel to the sides. The number of parallelograms thus formed is
1) $\frac{m^2}{4}$ 2) $\frac{(m+1)^2}{4}$
3) $\frac{(m+2)^2}{4}$ 4) $\frac{(m+1)^2 (m+2)^2}{4}$

23. The number of rectangles on a chess board is
1) 1296 2) 204 3) 1292 4) 200
24. The number of ways of choosing 2 squares (1×1) from a chess board so that they have exactly one common corner is
1) 98 2) 112 3) 36 4) 72
25. If t_n denotes the number of triangles formed with n points in a plane no three of which are collinear and if $t_{n+1} - t_n = 36$, then n = [EAM-2013]
1) 7 2) 8 3) 9 4) 10
26. A polygon has 54 diagonals. Then the number of its sides is [EAM-2010]
1) 7 2) 9 3) 10 4) 12
27. Note the arrangements (1, 1), (1, 2), (1, 3), (2, 3), (3, 3), (3, 4), (4, 4). Have we start from (1, 1), then increase one of the coordinates by 1 and repeat the same until we reach (4, 4). For example (1, 1), (2, 1), (2, 2), (2, 3), (3, 3), (3, 4), (4, 4) is another such arrangements. The number of such arrangements is
1) 6P_6 2) 6C_6 3) $\frac{6!}{3!3!}$ 4) 6

TOTAL NUMBER OF COMBINATIONS

28. The number of ways of selecting atleast one red ball from a bag containing 4 identical red balls and 5 identical black balls is
1) 20 2) 21 3) 23 4) 24
29. Given 5 different green, 4 different blue, and 3 different red dice. The number of combinations of dice that can be chosen by taking atleast one green and one blue die is
1) 3680 2) 3690 3) 3700 4) 3720
30. A zoo has 20 zebras, 12 Giraffes, 11 lions and 3 tigers. The number of ways a tourist can visit the animals so that he must see at least one tiger is
1) $21 \times 13 \times 12 \times 3$ 2) 7×2^{43}
3) $7 \times 21 \times 13 \times 12 - 1$ 4) 6×2^{43}
31. At an election a voter may vote for any number of candidates not greater than the number to be choosen. There are 10 candidates and 5 members are to be chosen. The number of ways in which a voter may vote is
1) 630 2) 632 3) 637 4) 640

DISTRIBUTION OF DISSIMILAR THINGS INTO GROUPS

32. At an election 3 wards of a town are canvassed by 4, 5 and 3 men respectively. If there are 20 men volunteering the number of ways they can be allotted to the different wards is

1) $\frac{20!}{3!4!5!}$ 2) $\frac{12!}{3!4!5!}$ 3) $\frac{20!}{3!4!5!8!}$ 4) $\frac{12!}{3!4!5!8!}$

33. The number of ways in which 12 balls can be divided between two friends, one receiving 8 and the other 4, is

1) $\frac{12!}{8!4!}$ 2) $\frac{12!2!}{8!4!}$ 3) $\frac{12!}{8!4!2!}$ 4) $\frac{12!}{4!}$

34. The number of ways of dividing 15 books into 3 groups of 3, 4, 8 books respectively is

1) $\frac{15!}{2!3!4!8!}$ 2) $\frac{15!}{(3!)^2 4!8!}$
3) $\frac{15!}{4!8!}$ 4) ${}^{15}C_3 \cdot {}^{12}C_4 \cdot {}^8C_8$

DISTRIBUTION OF SIMILAR THINGS INTO GROUPS

35. The number of ways in which 13 gold coins can be distributed among three persons such that each one gets at least two gold coins is

1) 36 2) 24 3) 12 4) 6

NUMBER OF DIVISORS

36. The number of positive divisors of $2^5 3^6 7^3$ is

1) 14 2) 167 3) 168 4) 210

37. Number of even divisors of 1600 is

1) 21 2) 18 3) 3 4) 6

APPLICATIONS OF ONTO FUNCTION

38. The no. of 5 digit numbers that can be made using the digits 1 and 2 and in which atleast one digit is different is

1) 31 2) 32 3) 30 4) 29

LEVEL-I (H.W.) - KEY

1)1 2)3 3)3 4)4 5)4 6)3 7)4
8)3 9)3 10)3 11)1 12)2 13)2 14)2
15)4 16)1 17)1 18)1 19)1 20)4 21)1
22)4 23)1 24)1 25)3 26)4 27)1 28)4
29)4 30)2 31)3 32)3 33)2 34)4 35)1
36)3 37)2 38)3

LEVEL-I (H.W.) - HINTS

1. ${}^nC_r = {}^nC_s \Rightarrow n = r + s = 3 + 9 = 12 ; {}^{12}C_2 = 66$

2. Use ${}^nC_r = \frac{n!}{(n-r)!r!}$

3. $\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r} = \frac{84}{36} = \frac{7}{3}$

$\frac{{}^nC_{r+1}}{{}^nC_r} = \frac{n-r}{r+1} = \frac{126}{84} = \frac{3}{2}, (n, r) = (9, 3)$

4. ${}^nC_5 + {}^nC_6 = {}^{(n+1)}C_6 > {}^{(n+1)}C_5 \Rightarrow n+1 > 6+5$
 $\Rightarrow n = 11(\text{least})$

5. ${}^nC_r + \sum_{i=1}^m {}^{(n+m-i)}C_{r-1} = {}^{(n+m)}C_r = {}^{26}C_5$

6. (synopsis)
 $n \cdot {}^{(n-1)}C_{r-1} = r \cdot {}^nC_r$

7. ${}^6C_3 \times {}^5C_2 + {}^6C_4 \times {}^5C_1 = 275$

8. ASIAN'S (6) EUROPEANS (3) AMERICANS (4)

1	1	2
1	2	1
2	1	1

${}^6C_1 \times {}^3C_1 \times {}^4C_2 + {}^6C_1 \times {}^3C_2 \times {}^4C_1$
 $+ {}^6C_2 \times {}^3C_1 \times {}^4C_1$
 $= 108 + 72 + 180 = 360$

9. ${}^nC_2 = 153 \Rightarrow n = 18$

10. Let n be no. of men $\Rightarrow$ total participants = n+2

$2({}^nC_2) - 2({}^2C_1 \cdot {}^nC_1) = 66 \Rightarrow n = 11$

$\Rightarrow n+2 = 13$

11. $12 - (A+B+C+D) = 8$

$x \rightarrow {}^8C_4 (2+4)! y \rightarrow {}^4C_4 (2+4)! \Rightarrow {}^8C_4 \cdot 6! \cdot 6!$

12. Total no. of selections - (no. of selections in which 3 objects are together or 2 objects are together)

$= {}^nC_3 - n - n \cdot (n-4) = \frac{n(n-4)(n-5)}{3!}$

13. Men (7) Women (4) Committee (5)

3 2 ${}^7C_3 \cdot {}^4C_2$

2 3 ${}^7C_2 \cdot {}^4C_3$

1 4 ${}^7C_1 \cdot {}^4C_4$

14. $\frac{2W; 3B; 4R}{{}^6C_2 \cdot {}^3C_1 + {}^6C_1 \cdot {}^3C_2 + {}^6C_0 \cdot {}^3C_3}$
15. ${}^8C_6 = {}^8C_2 = 28$
16. ${}^8C_7 = 8$
17. ${}^{(n-1)}C_2 = \frac{20}{100}({}^nC_3) \Rightarrow n = 15$
18. Between any two intersecting lines an angles formed in ${}^{15}C_2$
19. ${}^5C_3 - 3$. since 2, 3, 5; 2, 4, 6; 2, 3, 6 do not form a triangle.
20. Synopsis: $\frac{1}{6}n(n-4)(n-5) = \frac{1}{6} \cdot 8 \cdot 4 \cdot 3 = 16$
21. $\frac{n(n-3)}{2} = 275 \Rightarrow n(n-3) = 550 = 25 \times 22$
22. ${}^{(m+2)}C_2 \times {}^{(m+2)}C_2 = \frac{(m+2)^2(m+1)^2}{4}$
23. ${}^9C_2 \cdot {}^9C_2$
24. $7^2(2)$
25. $t_{n+1} - t_n = 36 \Rightarrow {}^{n+1}C_3 - {}^nC_3 = 36$
 ${}^nC_2 + {}^nC_3 - {}^nC_3 = 36$
 $\Rightarrow {}^nC_2 = 36 \Rightarrow {}^nC_2 = {}^9C_2 \Rightarrow n = 9$
26. If the number of sides of a polygon is n then the number of 4 diagonals $= n \frac{(n-3)}{2}$
 $\therefore \frac{n(n-3)}{2} = 54 \Rightarrow n(n-3) = 108 = 12 \times 9 \Rightarrow n = 12$
27. From a point (x_1, y_1) to reach the point (x_2, y_2)
 no. of ways $= {}^{(x_2-x_1+y_2-y_1)}C_{(x_2-x_1)} = {}^6C_3$
28. $(p+1)q = (5+1)4 = 24$
29. $(2^5 - 1)(2^4 - 1) \cdot 2^3$
30. $2^{20} \times 2^{12} \times 2^{11}(2^3 - 1)$
31. ${}^{10}C_1 + \dots + {}^{10}C_5 = 637$
32. ${}^{20}C_4 \cdot {}^{16}C_5 \cdot {}^{11}C_3 = \frac{20!}{3!4!5!8!}$
33. $2! \frac{12!}{8!4!}$ Interchange

34. $\frac{15!}{3!4!8!} = \frac{15!}{12!3!} \frac{12!}{4!8!} \frac{8!}{8!(8-8)!} = {}^{15}C_3 \cdot {}^{12}C_4 \cdot {}^8C_8$
35. After giving 2 coins each we have $13 - 3 \times 2 = 7$ coins. No. of non negative integral solution of $x_1 + x_2 + x_3 = 7$ is ${}^{(7+3-1)}C_{3-1} = {}^9C_2 = 36$
36. $(5+1)(6+1)(3+1) = 168$
37. $1600 = 2^6 \times 5^2$.
 No. of even divisors $= 6(2+1) = 18$
38. $2^5 - 2 = 30$



LEVEL-II (C.W)

SELECTION OF DISSIMILAR THINGS

- A father with 8 children takes them 3 at a time to the zoological garden, as often as he can without taking the same 3 children together more than once. Then the number of times a particular child will not go to the Zoological garden is
 1) 56 2) 21 3) 18 4) 35
- A party of 9 persons are to travel in two vehicles, one of which will not hold more than 7 and the other not more than 4. The number of ways the party can travel is
 1) 120 2) 220 3) 236 4) 246
- Out of 9 boys the number to be taken to form a group, so that the number of different groups may be greatest is
 1) 4 2) 5 3) 4 or 5 4) 6
- A train going from Vijayawada to Hyderabad stops at nine intermediate stations. Six persons enter the train during the journey with six different tickets of the same class. The no. of different tickets they may have
 1) ${}^{11}C_6$ 2) ${}^{45}C_6$ 3) 9C_6 4) ${}^{10}C_6$

COMMITTEE

- A committee of 5 men and 3 women is to be formed out of 7 men and 6 women. If two particular women are not to be together in the committee, the number of committees formed is
 1) 420 2) 5040
 3) 336 4) 216

6. A reserve of 12 railway station masters is to be divided into two groups of 6 each one for day duty and the other for night duty. The number of ways in which this can be done if two specified persons A, B should not be included in the same group is
1) 500 2) 504 3) 508 4) 512
7. A committee of 6 is chosen from 10 men and 7 women so as to contain atleast 3 men and 2 women. If 2 particular women refuse to serve on the same committee, the number of ways of forming the committee is
1) 7800 2) 8610 3) 810 4) 8000
8. The crew of an 8 oar boat is to be chosen from 12 Men, of whom 3 can row on stroke side only. If selected the number of ways the crew can be arranged is
1) ${}^9C_4 \cdot {}^8C_4$ 2) $14 \times 9!$
3) ${}^{12}C_8 \cdot {}^9C_4 \cdot 4! \cdot 4!$ 4) ${}^{12}C_4 \cdot {}^8C_4 \cdot 4! \cdot 4!$
9. A car will hold 2 persons in the front seat and 1 in the rear seat. If among 6 persons only 2 can drive, the number of ways, in which the car can be filled is
1) 10 2) 18 3) 20 4) 40

GEOMETRICAL APPLICATIONS

10. There are 'm' points on a straight line AB and 'n' points on another straight line AC in which A is not included. By joining these points triangles are constructed. i) When A is not included. ii) When A is included, the ratio of number of triangles in both cases is
1) $\frac{m+n-2}{m+n}$ 2) $\frac{m+n-2}{2}$
3) $\frac{m+n-2}{m+n+2}$ 4) $\frac{m+n+2}{m+n-2}$
11. In a polygon no three diagonals are concurrent. If the total number of points of intersection of diagonals interior to the polygon is 35 and the number of diagonals is 'x', number of sides is 'y' then (y, x) =
1) (5, 5) 2) (6, 9) 3) (5, 20) 4) (7, 14)
12. Let T_n denote the number of triangles which can be formed using the vertices of a regular polygon of n sides. If $T_{n+1} - T_n = 10$, then the value of n is [AIEEE-2013]
1) 5 2) 10 3) 8 4) 7

13. There are three coplanar lines, if any m points taken on each of the lines, the maximum number of triangle with vertices at these points is
1) $m^2(4m-3)$ 2) $3m^2(m-3)+1$
3) $3m^2(m-3)$ 4) $m(4m-3)$
14. ABCD is a convex quadrilateral 3, 4, 5 and 6 points are marked on the sides AB, BC, CD and DA respectively, the number of triangles with vertices on different sides is
1) 270 2) 220 3) 282 4) 342
15. The greatest number of points of intersection of 8 lines and 4 circles is
1) 64 2) 92 3) 104 4) 96

DISTRIBUTION OF DISSIMILAR THINGS

16. A guard of 15 men is formed form a group of n soldiers. The number of times two particular soldiers will guards
1) $\frac{n!}{13!2!}$ 2) $\frac{(n-2)!}{13!}$
3) $\frac{(n-2)!}{13!(n-15)!}$ 4) $\frac{(n-2)!}{13!12!}$
17. 15 Passengers are to travel by a double decker bus which can accomodate 5 in upper deck and 10 in lower deck. The number of ways that the passengers are distributed is
1) 3000 2) 3003 3) 3006 4) 3009

DISTRIBUTION OF SIMILAR THINGS

18. The number of ways of distributing 9 identical balls in 3 distinct boxes so that none of the boxes is empty is
1) 8C_3 2) 28 3) 3^8 4) 5
19. Five distinct letters are to be transmitted through a communication channel. A total number of 15 blanks is to be inserted between the the letters with at least three between every two. The number of ways in which this can be done is
1) 1200 2) 1800
3) 2400 4) 3000

20. The number of nonnegative integer solutions of the equation $x + y + z + 5t = 15$ is
 1) 196 2) 224
 3) 312 4) 364
21. The number of integral solutions to the system of equations $x_1 + x_2 + x_3 + x_4 + x_5 = 20$ and $x_1 + x_2 = 15$ when $x_k \geq 0$ ($k = 1, 2, 3, 4, 5$) is
 1) 300 2) 350
 3) 336 4) 316
22. If N is the number of positive integral solutions of $x_1 x_2 x_3 x_4 = 770$, then $N =$
 1) 256 2) 729
 3) 900 4) 770

SUM OF DIVISORS

23. The sum of divisors of $2^5 \cdot 3^4$ is
 1) $\frac{2^5-1}{2-1} \cdot \frac{3^4-1}{3-1}$ 2) $\frac{2^6-1}{2-1} \cdot \frac{3^5-1}{3-1}$
 3) $\frac{2^4-1}{2-1} \cdot \frac{3^3-1}{3-1}$ 4) $2^5 \cdot 3^4$
24. The sum of the even divisors of 168 is
 1) 448 2) 460 3) 42 4) 122
25. The sum of odd divisors of 360 is
 1) 70 2) 78 3) 80 4) 88
26. If N denotes set of all positive integers and if $f: N \rightarrow N$ is defined by $f(n) =$ the sum of positive divisors of n . then $f(2^k \cdot 3)$ where ' k ' is a positive integer is
 1) $2^{k+1} - 1$ 2) $2(2^{k+1} - 1)$
 3) $3(2^{k+1} - 1)$ 4) $4(2^{k+1} - 1)$

APPLICATIONS OF MULTINOMIAL THEOREM

27. A man goes in for an examination in which there are 4 papers with a maximum of 10 marks for each paper. The number of ways of getting 20 marks on the whole is
 1) 750 2) 860 3) 881 4) 891
28. How many different words can be formed out of the letters of the word "MORADABAD" taken 4 at a time
 1) 620 2) 622 3) 626 4) 624

MISCELLANEOUS PROBLEMS

29. The no. of permutations of the letters of the word 'PROPORTION' taken 4 at a time so that 3 are alike and one is different is
 1) 15 2) 20 3) 25 4) 120
30. The number $(36)!$ is divisible by
 1) 2^{36} 2) 2^{38} 3) $(12)^{17}$ 4) $(24)^{12}$
31. The number of divisors of 3630, which have a remainder of 1 when divided by 4, is
 1) 12 2) 6 3) 4 4) 18
32. If n is an integer with $0 \leq n \leq 11$ then the minimum value of $n!(11-n)!$ is attained when a value of $n =$ {EAMCET - 2014}
 1) 5 2) 7 3) 9 4) 11

LEVEL-II (C.W.) - KEY

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 1) 4 | 2) 4 | 3) 3 | 4) 2 | 5) 3 | 6) 2 | 7) 1 |
| 8) 2 | 9) 4 | 10) 1 | 11) 4 | 12) 1 | 13) 1 | 14) 4 |
| 15) 3 | 16) 3 | 17) 2 | 18) 2 | 19) 3 | 20) 2 | 21) 3 |
| 22) 1 | 23) 2 | 24) 1 | 25) 2 | 26) 4 | 27) 4 | 28) 3 |
| 29) 2 | 30) 3 | 31) 2 | 32) 1 | | | |

LEVEL-II (C.W.) - HINTS

1. No. of times he will not go to garden
 = No. ways of selecting 3 children from 7 = 7C_3
2. $\frac{A}{7} \frac{B}{2} \quad {}^9C_7 + {}^9C_6 + {}^9C_5 \quad (\text{or})$
 $\frac{6}{5} \frac{3}{4} \quad {}^9C_2 + {}^9C_3 + {}^9C_4$
3. 9C_r is greatest if $r = 4$ or 5
4. $(9+8+7+\dots+1)C_6$
5. ${}^7C_5 \cdot {}^6C_3 - {}^7C_5 \cdot {}^4C_1 = 336$
6. $2 \times {}^{12-2}C_{6-1}$
7. $({}^{10}C_3 \cdot {}^7C_3 + {}^{10}C_4 \cdot {}^7C_2) - ({}^{10}C_4 \cdot {}^5C_0 + {}^{10}C_3 \cdot {}^5C_1)$
8. Without the 3 - specified numbers for the non-stroke side = ${}^9C_4 \cdot 4!$. From remaining 8 members for stroke side = ${}^8C_4 \cdot 4!$
 Total = ${}^9C_4 \cdot 4! \times {}^8C_4 \cdot 4!$
9. No. Drivers = 2
 The drivers seat can be filled in 2C_1 ways. i.e., 2 ways out of the remaining persons = $4+1=5$.
 5 persons arrange in vacant seats in 5P_2 ways.
 $\therefore$ Total number of ways = $2 \times {}^5P_2 = 40$

10. i) When A is not included, the number of triangles formed = ${}^m C_2 \cdot {}^n C_1 + {}^m C_1 \cdot {}^n C_2$
 ii) When A is included, the number of triangles formed = ${}^m C_2 \cdot {}^n C_1 + {}^m C_1 \cdot {}^n C_2 + {}^m C_1 \cdot {}^n C_1$
11. ${}^n C_4 = 35 \Rightarrow n = 7$
 No. of diagonals = $\frac{n(n-3)}{2} = 14$
12. ${}^{(n+1)} C_3 - {}^n C_3 = 10 \Rightarrow n = 5$
13. Total triangles formed - No. of missing triangles
 = ${}^{(3m)} C_3 - 3 \cdot {}^m C_3 = m^2 (4m - 3)$
14. No. of triangles = $3 \times 4 \times 5 + 4 \times 5 \times 6 + 5 \times 6 \times 3 + 6 \times 3 \times 4 = 342$
15. $2 \cdot {}^4 C_2 + {}^8 C_2 + 2 \times 4 \times 8 = 12 + 28 + 64 = 104$
16. ${}^{n-2} C_{15-2}$
17. ${}^{15} C_5$ or ${}^{15} C_{10}$
18. The problem is equivalent to distributing 9 identical things to 3 persons so that each person can get at least one thing i.e. Number of positive integral solutions of $x_1 + x_2 + x_3 = 9 \Rightarrow {}^8 C_2 = 28$
19. For $1 \leq i \leq 4$ let $x_i (\geq 3)$ be the number of blanks between i^{th} and $(i+1)^{th}$ letter. Then $x_1 + x_2 + x_3 + x_4 = 15$. No. of solutions is coeff of p^{15} in $(p^3 + p^4 + \dots)^4 = {}^6 C_3 = 20$
 but 5 letters can be arranged in $5!$ ways
 $\therefore$ req. no. of arrangements = $20 \times 5! = 2400$
20. $x + y + z = 15 - 15t, t = 0, 1, 2, 3$
 The number of nonnegative integer solutions is

$$\binom{15-5t+3-1}{3-1} = \binom{17-5t}{2}$$

$$= \frac{(17-5t)(16-5t)}{2}$$

 The desired number is $\sum_{t=0}^3 \frac{(17-5t)(16-5t)}{2}$
 $= 136 + 66 + 21 + 1 = 224$
21. $x_1 + x_2 = 15; x_r \geq 0; r = 1, 2; n = 15$
 No. of non negative integral solutions
 $= {}^{n+r-1} C_{r-1} = {}^{16} C_1 = 16$
 $x_1 + x_2 + x_3 + x_4 + x_5 = 20$
 $x_3 + x_4 + x_5 = 5 \Rightarrow {}^{5+3-1} C_3 = {}^7 C_2 = 21$
 Total no. of solutions = $16 \times 21 = 336$

22. $770 = 2 \times 5 \times 7 \times 11$. We can assign 2 to x_1 or x_2 or x_3 or x_4 . Thus 2 can be assigned in 4 ways similarly each of 5, 7, 11 can be assigned in 4 ways.
 $\therefore$ No. of solutions = $4 \times 4 \times 4 \times 4 = 256$
23. $\left(\frac{2^{5+1}-1}{2-1} \right) \left(\frac{3^{4+1}-1}{3-1} \right)$
24. Sum of even divisors = sum of divisors - sum of odd divisors = $480 - 32 = 448$
25. $360 = 2^3 \times 3^2 \times 5^1$
 Sum of odd divisors = $\left(\frac{3^{2+1}-1}{3-1} \right) \left(\frac{5^{1+1}-1}{5-1} \right) = 78$
26. Sum of positive divisors
 $= \frac{(2^{K+1}-1)}{(2-1)} \times \frac{(3^2-1)}{(3-1)} = 4(2^{K+1}-1)$
27. The desired number is the coefficient of x^{20} in
 $(1+x+x^2+\dots+x^{10})^4 = (1-x^{11})^4 (1-x)^{-4}$
 $= (1-4x^{11}+\dots) \left(1 + \binom{4}{1}x + \binom{5}{2}x^2 + \dots \right)$
 which is $\binom{23}{20} - 4\binom{12}{9} = \binom{23}{3} - 4\binom{12}{3}$
 $= \frac{23 \cdot 22 \cdot 21 - 4 \cdot 12 \cdot 11 \cdot 10}{6} = 11(161 - 80) = 891$
28. M O R B A D
 A D
 A
 4 diff: ${}^6 P_4 = 360$
 2 diff + 2 same: ${}^2 C_1 \cdot {}^5 C_2 \cdot \frac{4!}{2!} = 240$
 3 same + 1 diff: ${}^5 C_1 \cdot \frac{4!}{3!} = 20$
 2 same + 2 same: $\frac{4!}{2! \cdot 2!} = 6$
 Total ways : 626
29. 3 alike = 3 O's = 1
 1 different from {P, R, T, I, N} = ${}^5 C_1$
 ${}^5 C_1 \cdot 1 \cdot \frac{4!}{3!}$

30. Exponent of 2 in $36!$ is

$$\left[\frac{36}{2}\right] + \left[\frac{36}{4}\right] + \left[\frac{36}{8}\right] + \left[\frac{36}{16}\right] + \left[\frac{36}{32}\right]$$

$$= 18 + 9 + 4 + 2 + 1$$

Exponent of 3 in $36!$ is

$$\left[\frac{36}{3}\right] + \left[\frac{36}{9}\right] + \left[\frac{36}{27}\right] = 12 + 4 + 1 = 17$$

$$36! = 2^{34} \times 3^{17} \times 5^{\alpha} \dots = 4^{17} \times 3^{17} \times 5^{\alpha} \dots$$

$36!$ is divisible by 12^{17}

31. Divisors are of the form $4m + 1$. i.e. they are odd divisors. Among odd divisors find the divisors which leave remainder 1 when divided by 4

32. Minimum of $n!(11-n)!$ is attained

$$\Rightarrow \text{Maximum of } \frac{11!}{n!(11-n)!} \text{ is attained}$$

$$\Rightarrow \text{Maximum of } {}^{11}C_n \text{ is attained } \Rightarrow n = 5 \text{ or } 6$$



LEVEL-II (H.W)

SELECTION OF DISSIMILAR THINGS

- A father with 6 children takes 3 at a time to a park without taking the same children. How often each child goes to the park ?
1) 10 2) 12 3) 15 4) 20
- A person wishes to make up as many different parties of 10 as he can out of 20 friends, each party consisting of the same number. In how many of the parties the same man is found ?
1) 380 2) 19 3) 90378 4) 92378
- ${}^nC_{r+1} + 2 {}^nC_r + {}^nC_{r-1} =$
1) ${}^{n+1}C_r$ 2) ${}^{n+2}C_r$ 3) ${}^{n+2}C_{r+1}$ 4) ${}^{n+2}C_{r+2}$

COMMITTEE

- From a council of 7 Hindus, 5 Muslims and 4 Christians a committee of 5 is to be formed with 1 christian and at least two Hindus. The number of ways of forming the committee is
1) 1220 2) 1240 3) 1680 4) 1280
- An examination paper, which is divided into two groups consisting of 3 and 4 questions respectively carries the note it is not necessary to answer all the questions. One question atleast should be answered from each group. The number of ways can an examinee select the questions is
1) 22 2) 105 3) ${}^3P_3 \times {}^4P_4$ 4) ${}^3C_3 \times {}^4C_4$

- A guard of 12 men is formed from n soldiers. If A and B are three times as often together on guard as C,D,E then =
1) 16 2) 32 3) 64 4) 8
- A question paper consisting of 10 questions is divided into 3 parts with 5, 3, 2 questions. A candidate is to answer 6 questions without neglecting any part. The number of ways in which he can make up his choice is
1) 175 2) 200 3) 225 4) 150

GEOMETRICAL APPLICATIONS

- The maximum number of points of intersection of 8 straight lines is
1) 8 2) 16 3) 28 4) 56
- The maximum number of points of intersection of 8 circles is
1) 16 2) 24 3) 28 4) 56
- There are 10 points in a plane and A is one of them. If no three of the points are collinear then the number of triangles formed with A as vertex is
1) 45 2) 36 3) 84 4) 120
- Six points are taken on a circle. The number of triangles formed inside the circle is
1) 20 2) 22 3) 25 4) 32

DISTRIBUTION OF DISSIMILAR THINGS

- The number of ways in which a pack of 52 cards of four different suits be distributed equally among 4 players so that each may have ace, king, queen and knave of the same suit is
1) $\frac{4!(36!)}{(9!)^4}$ 2) $\frac{36!}{(9!)^4}$ 3) $\frac{2(36!)}{(9!)^4}$ 4) $\frac{3(36!)}{(9!)^4}$
- A shop keeper sells three varieties of perfumes and he has a large number of bottles of the same size of each variety in this stock. there are 5 places in a row in his showcase. The number of different ways of displaying the three varieties of perfumes in the showcase is
1) 6 2) 50 3) 150 4) 60

DISTRIBUTION OF SIMILAR THINGS

- The number of ways of distributing 8 identical balls in three distinct boxes so that none of the boxes is empty
1) 5 2) 8C_3 3) 3^8 4) 21

15. The number of ways of selecting 8 books from a library which has 9 books each of Mathematics, Physics, Chemistry and English is

- 1) 165 2) ${}^{27}C_8$ 3) 3^8 4) 81

16. The number of ways in which an examiner can assign 30 marks to 8 questions giving not less than 2 marks to any questions is

- 1) ${}^{19}C_7$ 2) ${}^{20}C_7$ 3) ${}^{21}C_7$ 4) ${}^{22}C_7$

17. If a, b, c are three natural numbers in AP and $a + b + c = 21$ then the possible number of ordered triplets (a, b, c) is

- 1) 15 2) 14 3) 13 4) 12

18. There are 15 trees in a row. 4 trees are to be cut down. The number of ways that no two of the cut down trees are consecutive is

- 1) ${}^{11}C_4$ 2) ${}^{12}C_4$ 3) ${}^{15}C_4$ 4) ${}^{14}C_4$

MISCELLANEOUS PROBLEMS

19. If 7^r divides $55!$ then $r =$

- 1) 6 2) 7 3) 8 4) 9

20. Largest value of n , so that 10^n divides $51!$

- 1) 47 2) 12 3) 35 4) 59

21. The number of ways of writing 98 as the product of two positive integers is

- 1) 1 2) 2 3) 3 4) 4

22. Out of 5 apples, 10 mangoes and 15 oranges, the number of ways of distributing 15 fruits each to two persons is

- 1) 56 2) 64 3) 66 4) 72

23. The number of ways in which four letters can be selected from the letters of the word "MATHEMATICS" is

- 1) 133 2) 146 3) 136 4) 73

24. 3^r divides $50!$ then maximum value of $r =$

- 1) 27 2) 25 3) 23 4) 22

25. The number of ways in which 5 identical balls can be kept in 10 identical boxes, if not more than one can go into a box, is

- 1) ${}^{10}P_5$ 2) $\binom{10}{5}$ 3) 5 4) 1

LEVEL-II (H.W.) - KEY

- | | | | | | | |
|------|------|------|------|------|------|------|
| 1)1 | 2)4 | 3)3 | 4)3 | 5)2 | 6)2 | 7)1 |
| 8)3 | 9)4 | 10)2 | 11)1 | 12)1 | 13)3 | 14)4 |
| 15)1 | 16)3 | 17)3 | 18)2 | 19)3 | 20)2 | 21)3 |
| 22)3 | 23)3 | 24)4 | 25)4 | | | |

LEVEL-II (H.W.) - HINTS

- ${}^{n-s}C_{r-s} = {}^{6-1}C_{3-1} = {}^5C_2 = \frac{5 \times 4}{1 \times 2} = 10$
- ${}^{20-1}C_{10-1}$
- ${}^nC_{r+1} + {}^nC_r + {}^nC_r + {}^nC_{r-1}$
- ${}^4C_1 \cdot {}^7C_2 \cdot {}^5C_2 + {}^4C_1 \cdot {}^7C_3 \cdot {}^5C_1 + {}^4C_1 \cdot {}^7C_4 \cdot {}^5C_0$
- $(2^3 - 1)(2^4 - 1)$
- Include A, B in $12 = 3$ (include C, D, E)
 ${}^{n-2}C_{10} = 3 \quad {}^{n-3}C_9$
- ${}^5C_1 \cdot {}^3C_3 \cdot {}^2C_2 + {}^5C_3 \cdot {}^3C_1 \cdot {}^2C_2 + {}^5C_3 \cdot {}^3C_2 \cdot {}^2C_1$
 $+ {}^5C_2 \cdot {}^3C_3 \cdot {}^2C_1 + {}^5C_2 \cdot {}^3C_2 \cdot {}^2C_2 + {}^5C_4 \cdot {}^3C_1 \cdot {}^2C_1 = 175$
- ${}^8C_2 = 28$
- $2 \times {}^8C_2 = 2 \times 28 = 56$
- ${}^{10-1}C_{3-1} = 36$
- 6C_3
- From 4 suits $\rightarrow {}^4C_1$; 16 cards are distributed
 $\Rightarrow$ 36 are left over, 36 cards are distributed to 4 people = $\frac{4!(36)!}{(9!)^4}$
- Possibility selection arrangements
i) One triplet and 2 diff in = ${}^3C_1 \times \frac{5!}{3!} = 60$
ii) Two pairs and one diff. in = ${}^3C_2 \times \frac{5!}{2!2!} = 90$
 $\therefore$ Req. No. of ways = $60 + 90 = 150$
- ${}^{n-1}C_{r-1}$ where $n = 8, r = 3$
- Coefficient of x^8 in $(x^0 + x + \dots + x^9)^4$
i.e. No. of non negative integral solutions of
 $x_1 + x_2 + x_3 + x_4 = 8 \Rightarrow {}^{n+r-1}C_{r-1} = {}^{11}C_3 = 165$

16. $x_1 + x_2 + \dots + x_8 = 30$
 $X_i = x_i - 1 \Rightarrow X_1 + X_2 + \dots + X_8 = 22$
 The desired number is ${}^{21}C_7$
17. a, b, c are in AP $\Rightarrow 2b = a + c$
 $a + b + c = 21 \Rightarrow 3b = 21 \Rightarrow b = 7$
 $\therefore a + c = 14$
 No. of positive integral solutions = ${}^{13}C_1 = 13$
18. $n = 15, r = 4; {}^{(n-r+1)}C_r = {}^{12}C_4$
19. $r = \left[\frac{55}{7} \right] + \left[\frac{55}{7^2} \right] = 8$
20. The given question means, no. of zero's the end of
 $51!$ i.e. $\left[\frac{51}{2} \right] + \left[\frac{51}{2^2} \right] + \left[\frac{51}{2^3} \right] + \left[\frac{51}{2^4} \right] + \left[\frac{51}{2^5} \right] = 37$
 $\left[\frac{51}{5} \right] + \left[\frac{51}{5^2} \right] = 12$
 $\therefore 51!$ ends with $2^{12} \cdot 5^{12} = 10^{12}$ i.e. 12 zeros
21. $98 = 2^7 \cdot 7^2 \Rightarrow \frac{(1+1)(2+1)}{2} = 3$
22. The desired number is the coefficient of x^{15} in the product
 $(1 + x + x^2 + \dots + x^5)(1 + x + x^2 + \dots + x^{10})$
 $(1 + x + x^2 + \dots + x^{15})$
 $= (1 - x^6)(1 - x^{11})(1 - x^{16})(1 - x)^{-3}$
 $= (1 - x^6 - x^{11} + \dots) \left(1 + \binom{3}{1}x + \binom{4}{2}x^2 + \dots \right)$
 Which is $\binom{17}{15} - \binom{11}{9} - \binom{6}{4}$
 $= \binom{17}{2} - \binom{11}{2} - \binom{6}{2} = 136 - 55 - 15 = 66$
23. **MATHEMATICS**
 There are 2M's; 2A's 2T's and H,E,I,C,S
 We can select 4 letters from following cases
 (i) All different (ii) 2 similar; 2 similar
 (iii) 2 similar; 2 different
 ${}^8C_4 + {}^3C_2 + {}^3C_1 \cdot {}^7C_2 = 136$
24. $r = \left[\frac{50}{3} \right] + \left[\frac{50}{3^2} \right] + \left[\frac{50}{3^3} \right] = 16 + 5 + 1 = 22$
25. The 5 balls can be kept in 5 of the boxes in only one way



LEVEL - III

1. The number of rational numbers lying in the interval (2002, 2003) all of whose digits after the decimal point are non-zero and are in decreasing order is
 1) $\sum_{i=1}^9 {}^9P_i$ 2) 2^9 3) $2^9 - 1$ 4) $2^{10} - 1$
2. If the set $S = \{1, 2, 3, \dots, 12\}$ is to be partitioned into three sets A, B, C of equal size such that $A \cup B \cup C = S$, $A \cap B = B \cap C = A \cap C = \phi$ then the number of ways of partitioning S is
 1) $\frac{12!}{(3!)^4}$ 2) $\frac{12!}{3!(4!)^3}$ 3) $\frac{12!}{3!(3!)^4}$ 4) $\frac{12!}{(4!)^3}$
3. Two teams are to play a series of 5 matches between them. A match ends in a win or loss or draw for a team. A number of people forecast the result of each match and no two people make the same forecast for the series of matches. The smallest group of people in which one person forecast correctly for all the matches will contain 'n' people, where n =
 1) 81 2) 243 3) 486 4) 144
4. 18 guests have to be seated half on each side of a long table. 4 particular guests desire to sit on one particular side and 3 others on the other side. Then the number of ways in which the sitting arrangements can be made
 1) $(9!)^2$ 2) ${}^{11}C_4 (9!)^2$ 3) ${}^{11}C_3 (9!)^2$ 4) ${}^{11}C_5 (9!)^2$
5. If there are 5 periods in each working day of a school, then the number of ways that you can arrange 4 subjects during the working day is
 1) 220 2) 240 3) 260 4) 280
6. The number of positive integral solutions of $x^2 - y^2 = 352706$ is
 1) 2 2) 1 3) 0 4) 3
7. The number of different combinations that can be formed out of the letters of the word 'INFINITE' taken four at a time is
 1) 20 2) 22 3) 24 4) 120
8. Eight chairs are numbered 1 to 8. Two women and three men wish to occupy one chair each. First the women choose the chairs from among the chairs marked 1 to 4, then the men select the chairs from among the remaining. The number of possible arrangements is
 1) ${}^4P_2 \cdot {}^6P_3$ 2) ${}^6C_3 \cdot {}^4C_2$
 3) ${}^4C_2 \cdot {}^4P_3$ 4) ${}^4P_2 \cdot {}^4P_3$

9. There are 5 English, 4 Sanskrit and 3 Telugu books. Two books from each group are to be arranged in a shelf. The number of possible arrangements is
1) $(180) 6!$ 2) $(12) 7!$ 3) $7!$ 4) 180
10. The number of ways in which a TRUE or FALSE examination of n statements can be answered on the assumption that no two consecutive questions are answered the same way is
1) 2^{n-1} 2) 2^n 3) 1 4) 2
11. Three ladies have each brought their one child for admission to a school. The principal wants to interview the six persons one by one subject to the condition that no mother is interviewed before her child. The number of ways in which interviews can be arranged is
1) 6 2) 36 3) 72 4) 90
12. A delegation of four friends are to be selected from a group of 12 friends. The number of ways the delegation be selected if two particular friends refused to be together and two other particular friends wish to be together only in the delegation.
1) 226 2) 114 3) 156 4) 170
13. A class contains 4 boys and 'g' girls. Every Sunday five students, including at least three boys go for a picnic to Zoo Park, a different group being sent every week. During, the picnic, the class teacher gives each girl in the group a doll. If the total number of dolls distributed was 85, then value of 'g' is
1) 15 2) 12 3) 8 4) 5
14. The number of rectangles excluding squares from a rectangle of size 9×6 is
1) 391 2) 791 3) 842 4) 250
15. In the next world cup of cricket there will be 12 teams, divided equally in two groups. Teams of each group will play a match against each other. From each group 3 top teams will qualify for the next round. In this round each team will play against other once. Each of the four top teams of this round will play a match against the other three. Two top teams of this round will go to the final round, where they will play the best of three matches. The minimum number of matches in the next world cup will be
1) 54 2) 53 3) 38 4) 55
16. A boy has 3 library tickets and 8 books of his interest in the library. Out of these 8, he does not want to borrow Chemistry part II, unless Chemistry part I is also borrowed. The number of ways in which he can choose the three books to be borrowed is
1) 41 2) 32 3) 51 4) 26
17. Let $x, y, z = 105$ where $x, y, z \in \mathbb{N}$. Then number of ordered triplets (x, y, z) satisfying the given equation is
1) 15 2) 27 3) 6 4) 33
18. The no. of ways can a group of 5 letters be formed out of 5a 's, 5b 's, 5c 's and 5d 's is
1) ${}^5C_4 \times 5$ 2) ${}^5C_4 \times 5$ 3) 9C_5 4) 8C_5
19. A is a set containing n elements. A subset P of A is chosen. The set A is reconstructed by replacing the elements of P. A subset Q of A is again chosen. The number of ways of choosing P and Q so that $P \cap Q$ contains exactly two elements is
1) $9 \cdot {}^nC_2$ 2) $3^n - {}^nC_2$
3) $2 \cdot {}^nC_n$ 4) ${}^nC_2 \cdot 3^{n-2}$
20. There are 10 bags $B_1, B_2, B_3, \dots, B_{10}$ which contain 21, 22, ..., 30 different articles respectively. The total number of ways to bring out 10 articles from any one bag is
1) ${}^{31}C_{20} - {}^{21}C_{10}$ 2) ${}^{31}C_{21}$ 3) ${}^{31}C_{20}$ 4) ${}^{32}C_{20}$
21. A Tennis tournament is to be played by 10 pairs of students and each pair is to play with every other pair one set. If four sets are played each day then the number of days should be allowed for the tournament is
1) 12 2) 16 3) 80 4) 90
22. The results of 21 football matches (win, lose, draw) are to be predicted. The no. of different forecasts that can contain 19 wins is
1) 210 2) 640 3) 840 4) 1260
23. n bit strings are made by filling the digits 0 or 1. The number of strings in which there are exactly k zeros with no two 0's consecutive is
1) $(n-k)C_k$ 2) $(n-k+1)C_k$ 3) $(n-k-1)C_k$ 4) $(n-k)C_{(k-1)}$
24. The interior angles of a regular polygon measure 120° each. The number of diagonal of the polygon is
1) 8 2) 9 3) 18 4) 10

25. The number of ways in which we can select 4 numbers from 1 to 30 so as to exclude every selection of four consecutive numbers is
 1) 27378 2) 27405
 3) 27504 4) 27387
26. Given that 'n' is odd, the number of ways in which three numbers in A.P can be selected from 1, 2, 3, ..., n is
 1) $(n-1)^2/2$ 2) $(n+1)^2/4$
 3) $(n+1)^2/2$ 4) $(n-1)^2/4$
27. A child attempts to open a five disc-lock (each disc consists of digits {0, 1, ..., 9}) He takes 5 sec time to dial a particular number on the disc. If he does so for 5 hrs. every day, then the number of days he would be sure to open the lock is
 1) 30 2) 28 3) 27 4) 25
28. In a plane there are two families of lines $y = x + r$, $y = -x + r$, where $r \in \{0, 1, 2, 3, 4\}$. The number of squares of diagonals of the length 2 units formed by the lines is
 1) 9 2) 16 3) 25 4) 36
29. The number of ordered pairs (m, n) , $m, n \in \{1, 2, \dots, 100\}$ such that $7^m + 7^n$ is divisible by 5 is
 1) 1250 2) 2000 3) 2500 4) 5000
30. The number of different words which can be formed by taking 4 letters at a time out of the letters of the word 'EXPRESSION' is
 1) 2090 2) 2190
 3) 2454 4) 2354
31. The number of permutations of the letters of the word 'INDEPENDENT' taken 5 at a time is
 1) 3302 2) 3320 3) 3230 4) 3203
32. The number of ways that the letters of the word "PERSON" can be placed in the squares of the adjoining figure so that no row remains empty

$R_1 -$

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$R_2 -$

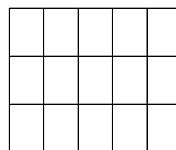
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$R_3 -$

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- 1) $20 \times 6!$ 2) $26 \times 6!$
 3) $20 \times 5!$ 4) $26 \times 5!$

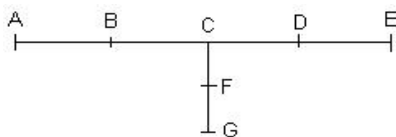
33. A rectangle with sides $2m-1$, $2n-1$ is divided into squares of unit length by drawing parallel lines as shown in the diagram.



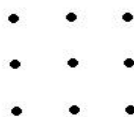
The number of rectangles with odd side length is

- 1) $(m+n+1)^2$ 2) $mn(m+1)(n+1)$
 3) m^2n^2 4) 4^{m+n-1}
34. The number of positive integral divisors of 1200 which are multiples of '6' is
 1) 6 2) 12 3) 8 4) 24
35. The sum of positive integral divisors of 600 which are multiples of 10 is
 1) 840 2) 3360 3) 1680 4) 420
36. The number of positive integral solution sets of the equation $xyz = 150$ is
 1) 27 2) 54 3) 108 4) 9
37. In how many ways 3 boys and 15 girls can sit together in a row such that between any 2 boys there is atleast 2 girls
 1) ${}^{14}C_3 3! 15!$ 2) ${}^{14}C_3 3! 10!$
 3) ${}^{18}C_3 3! 15!$ 4) ${}^{12}C_3 3! 15!$
38. A test has 4 parts. The first 3 parts carry 10 marks each and the 4th part carries 20 marks. Assuming that marks are not given in fractions, then the number of ways in which a candidate can get 30 marks out of 50 is
 1) 1011 2) 1111 3) 999 4) 1001
39. In an examination of 9 papers a candidate has to pass in more papers than the number of papers in which he fails in order to be successful. The number of ways in which he can be unsuccessful is
 1) 255 2) 256 3) 193 4) 319
40. The sum $S = \frac{1}{9!} + \frac{1}{3!7!} + \frac{1}{5!5!} + \frac{1}{7!3!} + \frac{1}{9!} =$
 1) $2^9/10!$ 2) $2^{10}/8$
 3) $2^{11}/9!$ 4) $\frac{2^{12}}{12!}$

41. The number of different triangles formed by joining the points A, B, C, D, E, F and G as shown in the figure given below is



- 1) 20 2) 24 3) 25 4) 30
42. Let A and B be two sets containing 2 elements and 4 elements respectively. The number of subsets of $A \times B$ having 3 or more elements is (EAM-2013)
- 1) 220 2) 219 3) 211 4) 256
43. How many different words can be formed by jumbling the letters in the word 'MISSISSIPPI' in which no two S are adjacent? (AIEEE-2008)
- 1) $7 \cdot {}^6C_4 \cdot {}^8C_4$ 2) $8 \cdot {}^6C_4 \cdot {}^7C_4$
 3) $6 \cdot 7 \cdot {}^8C_4$ 4) $6 \cdot 8 \cdot {}^7C_4$
44. The number of ways of selecting 3-member subset of $\{1, 2, 3, \dots, 25\}$ so that the numbers form a G.P. with integer common ratio is
- 1) 10 2) 11 3) 12 4) 15
45. The number of n digit numbers, which contain the digits 2 and 7, but not the digits 0, 1, 8, 9.
- 1) $6^n - 5^n - 5^n + 4^n$ 2) $6^n + 5^n - 5^n + 4^n$
 3) $6^n + 5^n - 5^n - 4^n$ 4) $6^n - 5^n - 4^n$
46. Let N be the number of 4 digit numbers formed with at most two distinct digits. Then the last digit of N is
- 1) 6 2) 7 3) 8 4) 9
47. If N is the least natural number which leaves remainders 2, 4, 6, 10 when divided by 3, 5, 7, 11 respectively, then the no. of divisors of N is
- 1) 4 2) 5 3) 6 4) 7
48. Nine points lie in a plane forming a square as shown



If N is the number of triangles with positive area, having three of these points as vertices, then the last digit of N is

- 1) 6 2) 7 3) 8 4) 9

49. Let n be any odd positive integer. The greatest integer k such that 2^k divides $n^{12} - n^8 - n^4 + 1$ is
- 1) 9 2) 10 3) 11 4) 12
50. The number of ways of selecting two 1×1 squares from a chess board such that they neither have a common vertex nor have a common side is N then last digit of N is
- 1) 3 2) 4 3) 5 4) 6
51. A_1, A_2, A_3, A_4, A_5 lie on L_1 . B_1, B_2, B_3, B_4, B_5 lie on L_2 and if L_1 is parallel to L_2 . If these points are joined and lines are formed, the maximum number of points of intersection that will form between the lines is
- 1) 80 2) 100 3) 120 4) 240
52. Let $f: A \rightarrow A$ be an invertible function where $A = \{1, 2, 3, 4, 5, 6\}$ The number of these functions in which at least three elements have self image is
- 1) 40 2) 56 3) 16 4) 3
53. The number of ways of dividing 15 men and 15 women into 15 couples, each consisting of a man and a woman is
- 1) 1240 2) 1840 3) 1820 4) 2005
54. The number of words of four letters containing equal number of vowels and consonants, repetition allowed is
- 1) 105^2 2) 210×243
 3) 105×243 4) 150×241
55. The number of ways of choosing 3 squares from a chess board so that they have exactly one common vertex
- 1) 195 2) 196 3) 197 4) 198
56. Number of squares of all dimensions of 5×7 game board
- 1) 70 2) 75 3) 80 4) 85
57. Maximum number of point of intersection made by 5 circles and 3 triangles
- 1) 70 2) 80 3) 128 4) 100

58. If there are five periods in each working day of a school, then the number of ways that you can arrange 3 subjects during the working day is
1) 125 2) 150 3) 160 4) 170
59. 8 points are marked on the circumference of a circle at equal distances. Then the squares can be drawn by joining them?
1) 8P_4 2) 8C_4 3) $\frac{{}^8C_4}{2}$ 4) 2
60. The number of ways in which one or more balls can be selected out of 10 white, 9 green and 7 black balls is (AIEEE - 2012)
1) 892 2) 881 3) 891 4) 879
61. If $s_n = \sum_{r=0}^n \frac{1}{n C_r}$ and $t_n = \sum_{r=0}^n \frac{r}{n C_r}$, then $\frac{t_n}{s_n}$
1) $\frac{1}{2}n$ 2) $\frac{2n-1}{2}$ 3) $n-1$ 4) $\frac{1}{2}n-1$
62. From 6 different novels and 3 different dictionaries, 4 novels and 1 dictionary are to be selected and arranged in a row on a shelf so that the dictionary is always in the middle. Then the number of such arrangements is (AIEEE-2009)
1) at least 500 but less than 750
2) at least 750 but less than 1000
3) at least 1000 4) less than 500

LEVEL-III KEY

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 1) 3 | 2) 4 | 3) 2 | 4) 4 | 5) 2 | 6) 3 | 7) 2 |
| 8) 1 | 9) 1 | 10) 4 | 11) 4 | 12) 1 | 13) 4 | 14) 2 |
| 15) 2 | 16) 1 | 17) 2 | 18) 4 | 19) 4 | 20) 1 | 21) 1 |
| 22) 3 | 23) 2 | 24) 2 | 25) 1 | 26) 4 | 27) 2 | 28) 1 |
| 29) 3 | 30) 2 | 31) 2 | 32) 2 | 33) 3 | 34) 2 | 35) 3 |
| 36) 2 | 37) 1 | 38) 2 | 39) 2 | 40) 1 | 41) 2 | 42) 2 |
| 43) 1 | 44) 1 | 45) 1 | 46) 1 | 47) 1 | 48) 1 | 49) 1 |
| 50) 4 | 51) 2 | 52) 2 | 53) 1 | 54) 2 | 55) 2 | 56) 4 |
| 57) 3 | 58) 2 | 59) 4 | 60) 4 | 61) 1 | 62) 3 | |

LEVEL - III - HINTS

1. A rational number of desired category is of the form 2002. $x_1x_2\ldots x_k$ where $1 \leq k \leq 9$ and $9 \geq x_1 > x_2 > \ldots > x_k \geq 1$ we can select K digits in 9C_k ways and arrange them in descending order in only one way.
 $\therefore$ No. of rational numbers is ${}^9C_1 + {}^9C_2 + \ldots + {}^9C_9 = 2^9 - 1$

2. From synopsis $\frac{12!}{(4!)^3}$
3. The smallest number of people = Total number of possible forecasts
= Total number of results (win, lose, draw)
= $3 \times 3 \times 3 \times 3 \times 3 = 3^5$
4. $18 - (4+3) = 11$; ${}^{11}C_5 (5+4)! {}^6C_6 \cdot (6+3)!$
5. ${}^4C_1 \cdot \frac{5!}{2!}$
(Select one subject, repeat it two times in 5 periods)
6. $(x^2 - y^2)$ is even
 $\therefore$ Both x and y are even or both odd
 $\Rightarrow (x+y)(x-y)$ is multiple of 4 which is not true as per data therefore number of solutions is zero
7. $3I + 2N + F + T + E$
3 same + 1 diff $\rightarrow 1 \times {}^4C_1$
2 same + 2 same $\rightarrow 1 \times 1$
2 same + 2 diff $\rightarrow 1 \times {}^4C_2 + 1 \times {}^4C_2$
all diff $\rightarrow {}^5C_4$
Total ${}^4C_1 + 1 + {}^4C_2 + {}^4C_2 + {}^5C_4 = 22$
8. Two women in 4 chairs and 3 men in remaining 6 chairs = ${}^4P_2 \cdot {}^6P_3$
9. ${}^5C_2 \cdot {}^4C_2 \cdot {}^3C_2 \cdot 6!$
10. TFTF..... - 1 Way; FTFT..... - 1 Way
Number of ways = 2
11. ${}^6C_2 \cdot {}^4C_2 \cdot {}^2C_2$
12. Let A, B wish to sit together and C, D do not wish to sit together. Then following cases arise.
i) A, B, C selected D not selected 8C_1
ii) A, B, D selected C not selected 8C_1
iii) A, B selected C, D not selected 8C_2
iv) D selected and A, B, C not selected 8C_3
v) C selected and A, B, D not selected 8C_3
vi) None of these are selected 8C_4
 $\therefore$ No. of ways = 226
13. Different groups formed
= ${}^4C_3 \times {}^8C_2 + {}^4C_4 \times {}^8C_1 = 4 \times {}^8C_2 + {}^8C_1$
No. of doll's distributed
= $2 \times 4 \times {}^8C_2 + g = 4g^2 - 3g = 85 \Rightarrow g = 5$
(There are two girls in each of $4 \times {}^8C_2$ groups)

14. Figure formed is 10×7 mutually perpendicular lines. No. of rectangles which are not squares is

$$\Rightarrow {}^{10}C_2 \times {}^7C_2 - \sum_{r=1}^6 (7-r)(10-r) = 791$$

15. No. of matches in first round = ${}^6C_2 + {}^6C_2$

$$\text{No. of matches in next round} = {}^6C_2$$

$$\text{No. of matches in Semi final round} = {}^4C_2$$

So required no. of matches

$${}^6C_2 + {}^6C_2 + {}^6C_2 + {}^4C_2 + 2 = 53 \quad (\text{minimum 2 matches must be played in best of 3})$$

16. If both chemistry books included (3rd book is selected in 6C_1) = 6

If chemistry II excluded then 3 books are selected in ${}^7C_3 = 35$ Total = $6 + 35 = 41$

17. $x \times y \times z = 105 \quad 3 \times 5 \times 7 = 105$

x can take any of the 3 numbers in 3 ways

y can take any of the 3 numbers in 3 ways

z can take any of the 3 numbers in 3 ways

$\therefore$ Total no. of solutions in $3 \times 3 \times 3 = 27$

18. By Synopsis ${}^{4+5-1}C_5$

19. (1) $a_1 \in P \& a_1 \in Q$ (2) $a_1 \in P \& a_1 \notin Q$

$$(3) a_1 \notin P \& a_1 \in Q \quad (4) a_1 \notin P \& a_1 \notin Q$$

$$A = \{a_1, a_2, \dots\} \quad \text{If } a_1, a_2 \in p \cap Q$$

Remaining (n-2) elements satisfy

$$(2) (3) (4) \text{ in } 3^{n-2} \text{ way } \therefore \text{Ans } {}^nC_2 \cdot 3^{n-2}$$

20. ${}^{21}C_{10} + {}^{22}C_{10} + {}^{23}C_{10} + \dots + {}^{30}C_{10} =$

$${}^{21}C_{11} + {}^{22}C_{12} + {}^{23}C_{13} + \dots + {}^{30}C_{20}$$

$$\left({}^nC_r = {}^nC_{n-r} \right) \text{ (add and subtract } {}^{21}C_{10} \text{)}$$

$${}^{21}C_{10} + {}^{21}C_{11} + {}^{22}C_{12} + {}^{23}C_{13} + \dots + {}^{30}C_{20} - {}^{21}C_{10}$$

$$= {}^{22}C_{11} + {}^{22}C_{12} + \dots + {}^{30}C_{20} - {}^{21}C_{10} = {}^{31}C_{20} - {}^{21}C_{10}$$

21. $\frac{\text{Number of games}}{4 \text{ sets}} = \frac{{}^{10}C_2}{4} = 12 \text{ days}$

22. ${}^{21}C_{19} \times 2 \times 2 = 840$

23. First arrange (n-k), 1's alternatively in 1 way. In (n-k+1) places arrange k zeros in ${}^{n-k+1}C_k$ ways.

24. $\left(\frac{2n-4}{n} \right) \times 90^0 = 120^0 \Rightarrow n = 6$

$$\text{no. of diagonals} = \frac{n(n-3)}{2} = 9.$$

25. ${}^{30}C_4 - 27 = 27378$

26. From synopsis

$$\left(\frac{n-1}{2} \right) C_2 + \left(\frac{n+1}{2} \right) C_2 = \left(\frac{n-1}{2} \right)^2$$

27. Each disc contains the digits 0, 1, 2, ..., 9

$$\text{Total no. of ways of dialing} = 10^5$$

$$\text{Time taken to dial each disc} = 5 \text{ sec} = \frac{5}{60 \times 60} \text{ hrs}$$

$$\therefore \text{Total time taken} = \frac{10^5 \times 5}{60 \times 60} \text{ hrs}$$

5 hrs of trial per 1 day

$$\therefore \frac{10^5 \times 5}{60 \times 60} \text{ hrs of trial per how many days?}$$

$$= \frac{10^5 \times 5}{60 \times 60 \times 5} = 27 \frac{7}{9} \text{ days i.e. 28 days}$$

28. Length of diagonal 2 units means length of side is

$\sqrt{2}$ units i.e. the two parallel lines must be alternate.

i.e. no. of ways = $3 \times 3 = 9$

29. Note that $7^r (r \in N)$ ends in 7, 9, 3 or 1 (corresponding to $r = 1, 2, 3$ & 4 respectively)

Thus, $7^m + 7^n$ cannot end in 5 for any values of $m, n \in N$. In other words, for $7^m + 7^n$ to be divisible by 5, it should end in 0.

For $7^m + 7^n$ to end in 0, the forms of m and n should be as follows.

	m	n
1	$4r$	$4s + 2$
2	$4r + 1$	$4s + 3$
3	$4r + 2$	$4s$
4	$4r + 3$	$4s + 1$

Thus, for a given value of m there are just 25 values of n for which $7^m + 7^n$ ends in 0.

[For instance, if $m = 4r$, then $n = 2, 6, 10, \dots, 98$]

$\therefore$ There are $100 \times 25 = 2500$ ordered pairs (m, n) for which $7^m + 7^n$ is divisible by 5.

30. SEXPRION
SE

$${}^8P_4 + {}^2C_1 \cdot {}^7C_2 \cdot \frac{4!}{2!} + {}^2C_2 \cdot \frac{4!}{2! \cdot 2!} = 2190$$

31. All different = ${}^6C_5 \cdot 5!$

$$2 \text{ similar} + 3 \text{ diff.} = {}^3C_1 \cdot {}^5C_3 \cdot \frac{5!}{2!}$$

$$3 \text{ same} + 2 \text{ diff.} = {}^2C_1 \cdot {}^5C_2 \cdot \frac{5!}{3!}$$

$$2 \text{ same} + 2 \text{ same} + 1 \text{ diff.} = {}^3C_2 \cdot {}^4C_1 \cdot \frac{5!}{2!2!}$$

$$3 \text{ same} + 2 \text{ same} = {}^2C_1 \cdot {}^2C_1 \cdot \frac{5!}{3!2!}$$

$\therefore$ Total selections = 72 Total arrangements = 3320

32. There are 6 different letters. We have to select 6 squares, taking atleast one from each row and then arrange in each selection.

R_1	R_2	R_3
(2)	(2)	(4)
1	1	4 $\rightarrow {}^2C_1 \cdot {}^2C_1 \cdot {}^4C_4 = 4$
1	2	3 $\rightarrow {}^2C_1 \cdot {}^2C_2 \cdot {}^4C_3 = 8$
2	1	3 $\rightarrow {}^2C_2 \cdot {}^2C_1 \cdot {}^4C_3 = 8$
2	2	2 $\rightarrow {}^2C_2 \cdot {}^2C_2 \cdot {}^4C_2 = 6$

Total number of selections of 6 squares = $4+8+8+6=26$. For each selection of 6 squares the number of arrangements of 6 letters = 6!

Total no. of ways = $26 \times 6! = 18720$

33. Along horizontal side one unit can be taken in $(2m-1)$ ways. 3 units can be taken in $(2m-3)$ ways and so on.... The number of ways of selecting a side horizontally is

$$(2m-1) + (2m-3) + (2m-5) + \dots + 3 + 1 = m^2$$

similarly the number of ways of selecting a side vertically is n^2 . Total number of rectangles = $m^2 n^2$.

34. For the number $2^a 3^b 5^c 7^d$, the number of positive integral divisors which are multiples of 6 is $ab(c+1)(d+1)$

35. For the number $2^a 3^b 5^c 7^d$, the sum of positive integral divisors which are multiples of 10

$$\text{is } \left(\frac{2^{a+1}-2}{2-1} \right) \left(\frac{3^{b+1}-1}{3-1} \right) \left(\frac{5^{c+1}-5}{5-1} \right) \left(\frac{7^{d+1}-1}{7-1} \right) \dots$$

36. $xyz = 150 = 2 \times 3 \times 5^2$ no. of positive integral solution sets = $3 \cdot 3(3_{C_1} + 3_{C_2})$

37. $x_1 + x_2 + x_3 + x_4 = 15$

subject to the conditions

$$x_1 \geq 0, x_2 \geq 2, x_3 \geq 2, x_4 \geq 0.$$

Now 3 boys can be permuted in their places in 3! ways and 15 girls can be permuted in 15! ways.

$$\text{So required number of ways} = {}^{14}C_3 (3!)(15!)$$

38. $x_1 + x_2 + x_3 + x_4 = 30$

where $0 \leq x_1, x_2, x_3 \leq 10$ and $0 \leq x_4 \leq 20$

= coefficient of x^{30} in the expression

$$(x^0 + x^1 + x^2 + \dots + x^{10})^3 (x^0 + x^1 + x^2 + \dots + x^{20})$$

$$= \text{coefficient of } x^{30} \text{ in } \left(\frac{1-x^{11}}{1-x} \right)^3 \left(\frac{1-x^{21}}{1-x} \right)$$

39. The candidate is unsuccessful if he fails in 9 or 8 or 7 or 6 or 5 papers. so that the number of ways to be unsuccessful

$$= {}^9C_9 + {}^9C_8 + {}^9C_7 + {}^9C_6 + {}^9C_5$$

$$= {}^9C_0 + {}^9C_1 + {}^9C_2 + {}^9C_3 + {}^9C_4$$

$$= \frac{1}{2} ({}^9C_0 + {}^9C_1 + \dots + {}^9C_9) = 2^8.$$

40.. We can write S as follows

$$S = \frac{1}{10!} [{}^{10}C_1 + {}^{10}C_3 + {}^{10}C_5 + {}^{10}C_7 + {}^{10}C_9] = \frac{1}{10!} (2^9)$$

41. No. of different triangles =

$${}^7C_3 - 5{}^5C_3 - 3{}^3C_3 = 24$$

42. $A \times B$ will have 8 elements

$$\text{Ans} = 2^8 - 8{}^8C_0 - 8{}^8C_1 = 256 - 1 - 8 - 28 = 219$$

43. Given word is **MISSISSIPPI**

Here, I=4 times, S=4 times

p=2 times M=1 time

 m I I I I P P

$$\therefore \text{Required number of words} = {}^8C_4 \times \frac{7!}{4!2!}$$

$$= {}^8C_4 \times \frac{7 \times 6!}{4!2!} = 7 \cdot {}^8C_4 \cdot {}^6C_4$$

44. 1,2,4; 2,4,8; 3,6,12; 4,8,16; 5,10,20; 6,12,24; 1,3,9; 2,6,18; 1,4,16; 1,5,25. The desired number is $6+2+1+1=10$

45. The total number of numbers without any restrictions containing digits 2,3,4,5,6,7, is $n(S) = 6^n$.

The total number of numbers that contain 3,4,5,6,7 is $n(A) = 5^n$.

The total number of numbers that contain 2,3,4,5,6 is $n(B) = 5^n$.

The total number of numbers that contain 3,4,5,6 is $n(A \cap B) = 4^n$.

The total number of numbers that do not contain digits 2 and 7 is $5^n + 5^n - 4^n$. The total number of numbers that contain 2 and 7 is

$$6^n - 5^n - 5^n + 4^n.$$

46. Let a and b be two non zero digits. Number of numbers using a and 0 is

$$\left(\binom{3}{0} + \binom{3}{1} + \binom{3}{2} + \binom{3}{3} \right) \times 9 = 8 \times 9 = 72$$

Like aaaa, aaa0, aa00, a000,

The number of numbers using a and b is

$$\left(\frac{4!}{1!3!} + \frac{4!}{(2!)^2} \right) \times 9 \times 8 = 7 \times 72$$

Like abbb, aabb, abbb ...

$$\therefore N = 72 + 7 \times 72 = 8 \times 72 = 576.$$

47. $N = 3.5.7.11 - 1 = 1154 = 2^1 \times 577^1$ which has $2 \times 2 = 4$ divisors.

48. The number of triangles with two dots in a row and other dot in a different row is $3 \times 3 \times 6 = 54$.

The number of triangles with two dots in a column and other dot is a different column is

$$3 \times 3 \times 4 = 36.$$

The number of triangles with each dot belonging to different row and different column is 4

$$\therefore N = 54 + 54 - 36 + 4 = 76.$$

49. $N = n^{12} - n^8 - n^4 + 1 = (n^8 - 1)(n^4 - 1)$
 $= (n^4 + 1)(n^4 - 1)^2 = (n^4 + 1)(n^2 + 1)^2 (n - 1)^2 (n + 1)^2$

As n is odd, $n^4 + 1, n^2 + 1, n + 1, n - 1$ are even

$\therefore N$ is divisible by 2^7

Further, one of $(n - 1)$ and $(n + 1)$ is divisible by 4.

$\therefore N$ is divisible by $2^{7+2} = 2^9$.

$$50. 64c_2 - (7 \times 7 \times 2) - (7 \times 8 \times 2) \\ = 32 \times 63 - 98 - 112 = 32 \times 63 - 210 = 1806$$

$$51. (5c_2)(5c_2)(1) = 100$$

$$52. {}^nC_r r! \left(\frac{1}{0!} - \frac{1}{1!} + \frac{1}{2!} - \dots \right)$$

Required number of functions

$$= {}^6C_3(2) + {}^6C_4(1) + {}^6C_5(0) + {}^6C_6$$

$$53. 15_{C_1} \times 15_{C_1} + 14_{C_1} \times 14_{C_1} + \dots + 1_{C_1} \times 1_{C_1} = \Sigma 15^2$$

54. The number of selections of 1 pair of vowels and 1

pair of consonants = ${}^5C_1 \times {}^{21}C_1$

$\therefore$ The required number of words

$$= {}^5C_1 \times {}^{21}C_1 \times \frac{4!}{2!2!} + {}^5C_2 \times {}^{21}C_2 \times 4!$$

$$55. 7^2(4)$$

$$56. (5 \times 7) + (4 \times 6) + (3 \times 5) + (2 \times 4) + (1 \times 3)$$

$$57. {}^5C_2(2) + {}^3C_2(6) + {}^5C_1 \cdot {}^3C_1(6)$$

$$58. {}^3C_2 \times \frac{5!}{2!2!} + {}^3C_1 \times \frac{5!}{3!}$$

59. Since 8 points divide the circumference into 8 equal parts, only 2 squares can be drawn.

60. Required number of ways =

$$= (10+1)(9+1)(7+1) - 1 = 880 - 1 = 879$$

$$61. t_n = \sum_{r=0}^n \frac{r}{{}^nC_r} = \sum_{r=0}^n \frac{n-r}{{}^nC_{n-r}}$$

$$= \sum_{r=0}^n \frac{n-r}{{}^nC_r} = n \sum_{r=0}^n \frac{1}{{}^nC_r} - \sum_{r=0}^n \frac{r}{{}^nC_r}$$

$$= n.s_n - t_n \Rightarrow 2t_n = ns_n \Rightarrow \frac{t_n}{s_n} = \frac{n}{2}$$

62. Out of 6 novels, 4 novels can be selected in 6C_4 ways. Also out of 3 dictionaries, 1 dictionary can be selected in 3C_1 ways. Since the dictionary is fixed in the middle, we only have to arrange 4 novels which can be done in $4!$ ways. Then the number of

$$\text{ways} = {}^6C_4 \cdot {}^3C_1 \cdot 4! = 6 \cdot \frac{5}{2} \cdot 3 \cdot 24 = 1080$$



LEVEL-IV

- Out of 7 men and 4 women a committee of 5 is to be formed. The number of ways in which this can be done
 I: so as to include exactly 2 women is 210.
 II: so as to include atleast 2 women is 301.
 Which of the above statements is true?
 1) only I 2) only II
 3) Both I and II 4) Neither I nor II
- I. The no. of ways of dividing 15 different objects into 3 equal groups is $\frac{15!}{5!5!5!}$
 II. The no. of ways in which 52 cards can be distributed among 4 persons equally is $\frac{52!}{(13!)^4 \cdot 4!}$
 Which of the above statement is correct?
 1) Only I 2) Only II
 3) Both I and II 4) Neither I nor II
- I : The no. of 3 digit numbers of the form xyz where $x > y > z$ is 1.
 II : The no. of 3 digit numbers of the form xyz where $x > y > z$ is ${}^{10}C_3 \cdot 3!$
 Which of the above statement is correct?
 1) only I 2) only II
 3) Both I and II 4) Neither I nor II
- In the shop there are five types of ice-creams available. A child buys six ice-creams.
 Statement - 1 : The number of different ways the child can buy the six ice - creams is ${}^{10}C_5$
 Statement - 2 : The number of different ways the child can buy the six ice-creams is equal to the number of different ways of arranging 6 A's and 4 B's in a row
 1) Statement-1 is true, Statement-2 is true and Statement - 2 is not a correct explanation for Statement - 1
 2) Statement-1 is true, Statement-2 is false
 3) Statement-1 is false, Statement-2 is true
 4) Statement-1 is true, Statement-2 is true and Statement-2 is a correct explanation for Statement-1

- Arrange the following values in ascending order.
 A : no. of divisors of 24
 B : no. of divisors of 12
 C : no. of divisors of 72
 D : no. of divisors of 120
 1) BACD 2) DCAB
 3) BADC 4) ABCD
- Arrange the following values in ascending order.
 A : No. of diagonals of a polygon with 10 sides
 B : No. of squares (exclusively squares) in a chess board
 C : No. of ways in which 4 boys and six girls sit alternately in a row
 D : No. of sides of the polygon in which no. of sides is equal to no. of diagonals.
 1) BADC 2) DCAB
 3) CDBA 4) CDAB
- Assertion(A): If a polygon has 35 diagonals then the number of the sides of the polygon is 15.
 Reason(R): The number of diagonals of a polygon with 'n' sides is $\frac{n(n-3)}{2}$
 1) Both A and R are true and R is the correct explanation of A
 2) Both A and R are true and R is not correct explanation of R
 3) A is true but R is false
 4) A is false but R is true.
- Assertion (A): The no. of parallelograms in a chess board is 1296.
 Reason(R): The no. of parallelograms when a set of 'm' parallel lines is intersected by another set of 'n' parallel lines is ${}^mC_2 \cdot {}^nC_2$
 1) A is true and R is false
 2) A is false and R is true
 3) Both A and R are true and R is correct explanation of A
 4) Both A and R are true and R is not correct explanation of A.

9. If given n points are on the circumference of a circle then observe the following Lists:

List - I

List - II

A. Number of straight lines i. $2^n - 1$

B. Number of diagonals in an n -sided closed polygon ii. nC_4

C. Number of triangles iii. ${}^nC_2 - n$

D. Number of quadrilaterals iv. nC_3

v. nC_2

The correct Match for List - I from List - II is

	A	B	C	D
1.	ii	i	iv	i
2.	v	iv	iii	i
3.	v	iii	iv	i
4.	i	iii	ii	iv

10. I: The total number of ways in which a selection (one or more) can be made out of $p + q + r$ things of which p are all alike, q all alike, r all alike is $(p+1)(q+1)(r+1) - 1$

II: The number of permutations of ' n ' things taken together when ' p ' of the things are alike of one kind, ' q ' of them alike of a second kind, ' r ' of them alike of a third kind and the rest all

different is $\frac{n!}{p!q!r!3!}$

Which of the above statement is true?

- 1) only I 2) only II
3) Both I and II 4) Neither I nor II

11. Statement -1: The number of ways of distributing 10 identical balls in 4 distinct boxes such that no box is empty is 9C_3

Statement-2: The number of ways of choosing any 3 places from 9 different places is 9C_3

(AIEEE- 2011)

- 1) Statement-1 is true, statement-2 is false
2) Statement-1 is false, Statement-2 is true.
3) statement-1 is true, statement-2 is true, statement-2 is a correct explanation for statement 1
4) Statement-1 is true, Statement-2 is true, statement-2 is not a correct explanation for statement-1

12. If a is the positive odd divisors of 216, b is the number of even divisors of 1600 and c is the number of odd divisors of 36000 then descending order of a, b, c is

- 1) a, b, c 2) b, c, a 3) c, a, b 4) a, c, b

13. Assertion (A): The number of positive divisors of $2^5 3^6 7^3$ is 168

Reason (R): The number of positive divisors of $x^n y^m z^r$ (here x, y and z are prime numbers) is $(x+3)(y+4)(z-4)$

1) Both A and R are true and R is the correct explanation of A

2) Both A and R are true and R is not correct explanation of R

3) A is true but R is false

4) A is false but R is true.

14. Observe the following Lists

List - I

List - II

- A. The number of ways of

answering one or more of n questions (i) $\frac{{}^nP_r}{2^r}$

- B. The number of ways of answering one or more of n questions when each question has an alternative is (ii) $2^n - 1$

- C. The number of circular permutations of n different

things taken r at a time is (iii) $\frac{{}^nP_r}{r}$

- D. The number of circular permutations of n things taken r at a time in one direction is (iv) $3^n - 1$
(v) 2^n

The correct match is

	A	B	C	D
1.	ii	iv	iii	i
2.	ii	iii	i	iv
3.	iii	ii	i	iv
4.	iv	iii	ii	i

PASSAGE : 1

Five balls are to be placed in three boxes. Each can hold all the five balls. The no. of ways of placing the balls so that no box remains empty

15. If balls and boxes are all different is

- 1) 50 2) 100 3) 125 4) 150

16. If balls are identical but boxes are different is

- 1) 2 2) 6 3) 4 4) 8

17. If balls are different but boxes are identical is

- 1) 25 2) 15 3) 10 4) 35

18. If balls as well as boxes are identical is

- 1) 1 2) 2 3) 25 4) 150

19. Let $S_1 = \sum_{j=1}^{10} j(j-1)^{10} C_j$, $S_2 = \sum_{j=1}^{10} j^{10} C_j$ and

$$S_3 = \sum_{j=1}^{10} j^2 \cdot {}^{10}C_j \quad (\text{AIEEE-2010})$$

Statement -1 : $S_3 = 55 \times 2^9$.

Statement -2 : $S_1 = 90 \times 2^8$ and $S_2 = 10 \times 2^8$

- 1) Statement-1 is true, Statement-2 is true, statement-2 is a correct explanation for statement 1
- 2) statement-1 is true, statement-2 is true, statement-2 is not a correct explanation for statement-1
- 3) Statement-1 is true, statement-2 is false
- 4) Statement-1 is false, Statement-2 is true.

LEVEL - IV - KEY

- 1) 3 2) 4 3) 4 4) 3 5) 1 6) 4 7) 4
8) 3 9) 3 10) 1 11) 4 12) 2 13) 3 14) 1
15) 4 16) 2 17) 1 18) 2 19) 3

LEVEL - IV - HINTS

1. I: ${}^7C_3 \times {}^4C_2 = 210 \Rightarrow \text{true}$
II: ${}^7C_3 \times {}^4C_2 + {}^7C_2 \times {}^4C_3 + {}^7C_1 \times {}^4C_4 = 301 \Rightarrow \text{true}$
2. I: $\frac{15!}{(5!)^3 3!} \Rightarrow \text{false}$ II: $\frac{52!}{(13!)^4} \Rightarrow \text{false}$
3. I: Select 3 digits from 10 digits $\{0, 1, 2, \dots, 9\}$ in ${}^{10}C_3$ ways and arrange as can be given. This is done in 1 way $\Rightarrow {}^{10}C_3$ (false) II: ${}^{10}C_3$ is true
4. $x_1 + x_2 + x_3 + x_4 + x_5 = 6$ $x_k \geq 0$
No. of solutions = ${}^{6+5-1}C_{5-1} = {}^{10}C_4 = \frac{10!}{6!4!}$
5. A: $24 = 2^3 \times 3^1 \Rightarrow 4 \times 2 = 8$
B: $12 = 2^2 \times 3^1 \Rightarrow 3 \times 2 = 6$
C: $72 = 2^3 \times 3^2 \Rightarrow 4 \times 3 = 12$
D: $120 = 2^3 \times 3^1 \times 5^1 \Rightarrow 4 \times 2 \times 2 = 16$
Ascending order BACD
6. A: $\frac{10(7)}{2} = 35$, B: $\sum_{n=1}^8 n^2 = 204$ C: 0 D: 5

Ascending order is CDAB

7. $\frac{n(n-3)}{2} = 35 \Rightarrow n(n-3) = 10 \cdot 7 \Rightarrow n = 10$
A is false & R is true
8. A: ${}^9C_2 \times {}^9C_2 = 1296$ - true
R is true and correct explanation of A
9. A: nC_2 , B: ${}^nC_2 - n$, C: nC_3 , D: nC_4
10. I: true II: $\frac{n!}{p!q!r!} \Rightarrow \text{false}$
11. The number of ways of distributing 10 identical balls in 4 distinct boxes such that no box is empty is ${}^{n-1}C_{r-1} = {}^{10-1}C_{4-1} = {}^9C_3$
The number of ways of choosing 3 places from 9 different places is ${}^nC_r = {}^9C_3$.
12. $216 = 2^3 \times 3^3$; No. of odd divisors = $(3+1) = 4$
 $1600 = 2^6 \times 5^2$; No. of even divisors = $6(2+1) = 18$
 $36000 = 2^5 \times 3^2 \times 5^3$
No. of odd divisors = $(2+1)(3+1) = 12$
13. No. of positive divisors = $6 \times 7 \times 4 = 168$ is true
R: $(n+1)(m+1)(r+1) \Rightarrow \text{false}$
14. A: ${}^nC_1 + \dots + {}^nC_n = 2^n - 1$
B: $3^n - 1$ C: $\frac{{}^nP_r}{r}$ D: $\frac{{}^nP_r}{2r}$
15. Number of ways $(3^5 - 3 \cdot 2^5 + 3) = 150$
16. $x_1 + x_2 + x_3 = 5$; $x_1, x_2, x_3 \geq 1$
Number of ways = ${}^4C_2 = 6$
17. Possibilities are $(3, 1, 1), (2, 2, 1)$ No. of ways = $\frac{{}^5C_2 \cdot {}^3C_2 \cdot {}^1C_1}{2!} + \frac{{}^5C_3 \cdot {}^2C_1 \cdot {}^1C_1}{2!} = (10 + 15) = 25$
18. Only two ways $(3, 1, 1)(2, 1, 1)$
19. $S_1 = \sum j(j-1) {}^{10}C_j = \sum j(j-1) \cdot \frac{10(10-1)}{(j-1)} \cdot {}^8C_{j-1} = 9 \times 10 \sum {}^8C_{j-2} = 90 \times 2^8$
 $S_2 = \sum j \cdot {}^{10}C_j = 10 \sum {}^9C_{j-1} = 10 \times 2^9$
 $S_3 = \sum j^2 \cdot {}^{10}C_j = \sum (j(j-1) + j) \cdot {}^{10}C_j$
 $= \sum j(j-1) {}^{10}C_j + \sum j \cdot {}^{10}C_j$
 $= 90 \cdot 2^8 + 10 \cdot 2^9 = (45 + 10) 2^9 = 55 \cdot 2^9$
Then statement -1 is true and statement-2 is false.

